

# PRA-MLX: Query-Addressed Memory beside Prompt and Rotating KV Caches

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## Abstract

Apple Silicon changes the physical cost model of long-context inference: weights, cache arrays, and host code share unified memory. It does not remove the logical question of which retained evidence should be active. We implement an MLX adapter boundary for Progressive Retrieval Attention (PRA), measure selected-text coexistence with the MLX-LM nearest-prefix cache, and test whether a rotating sequential cache can serve as a simpler archive. On an Apple M5 with the 4-bit Qwen3-0.6B checkpoint, selected and full context recover a fixed answer in 5/5 server requests; selected context uses 365 rather than 1,577 prompt tokens. Warm requests reuse 364 and 1,576 cached tokens, respectively, while the server prompt cache grows to 0.28 GB across five distinct sequences. In a separate five-seed Python-API experiment, full sequential KV recovers the answer in 5/5 cases with 112 MB of cache. Rotating KV at 64–512 tokens recovers 0/5. Reintroducing the selected archive record reaches 2/5 only at 64 tokens and 0/5 at larger capacities. We then implement native selected K/V at every layer. Across five seeds it is bit-exact to ordinary split-cache execution, recovers 5/5 answers, and remains 5/5 when local sequential K/V rotates at 64 tokens. Thus native PRA restores archived evidence without making it visible prompt text. A follow-up profile sweep finds that the last 14 of 28 layers retain 5/5 recovery with 7.71 rather than 15.42 MiB active K/V. Persisted K/V reloads in 7.2 ms with zero logit error, and shared-memory throughput rises from 6.04 to 16.01 requests/s between concurrency one and eight. On 40 QASPER and 40 HotpotQA natural-text transport controls per condition, ordinary and native memory rank the answer correctly in every case; disabled and shuffled memory remain near chance. A larger Qwen3-1.7B natural-QA study then uses the datasets’ original questions and answers. Full-precision native K/V exactly matches ordinary split-cache token F1 and answer log-probability on QASPER, HotpotQA, and 2WikiMultiHopQA. Shuffled and absent memory degrade both metrics on QASPER and HotpotQA; on 2Wiki the gold-answer log-probability degrades while free-generation F1 is noisy. Per-head int8 residency halves selected-K/V bytes and preserves this small cohort’s quality, with request-local dequantization before attention. An independent M4 Pro replication adds Qwen3-4B and preserves ordinary/native aggregate F1 exactly in all six model–dataset cohorts; shuffled-memory F1 falls to 0.011–0.028. A subsequent timing audit retires that runner’s apparent 44.5–54.5% native cost: MLX lazily completed ordinary source encoding inside the timed request while native encoding had already been synchronized. Finally, an SDK hybrid router selects four documents or paper paragraphs before native materialization. Native and ordinary consumption agree exactly on all 60 paired generations. Gold evidence recall at four is 0.348 on QASPER, 0.725 on HotpotQA, and 0.775 on 2Wiki. Routed memory outperforms shuffled memory on answer likelihood in all three datasets, but remains below oracle evidence; discovery, rather than MLX transport, is now the measured bottleneck. On the same frozen routed selections, a matched E0/E2 economic benchmark preserves all 840 output pairs across cold, warm, multi-query, and concurrency-eight schedules and removes 90.6–93.0% of visible prompt tokens. Native ingestion is cheaper than ordinary source-cache construction, but the unfused selected-cache path yields macro E2/E0 cost ratios of 1.130, 1.155, 1.148, and 1.132 across those schedules. An expanded 149-question confirmation preserves 2,086/2,086 pairs with ratios of 1.043, 1.139, 1.133, and 1.072. A 1,020-request pressure sweep then holds QA quality constant while varying an eight-resource session’s compact-K/V budget. Budgets below eight reload every repeated access; fitting all eight resources eliminates reloads and halves mean resolution time at a 151–168 MiB residency cost. An expanded live lifecycle study adds layer-segmented mmap WARM storage: WARM is 80/80 exact and restart recovery succeeds, while int8 COLD is exact in only 13/80 sequences and remains an explicit, unpromoted option. Five-example Llama and Gemma replications also preserve every lossless WARM output, while their small int8 results remain calibration observations rather than defaults. Event-loop promotion hides the WARM stall with sufficient lead, and the native streaming gateway measures queueing through concurrency sixteen with cancellation-safe cleanup. A 1,125-request long-session extension confirms an eight-resource working-set threshold, while selective int8 remains below the lossless gate. An additional 1,500-request Qwen3-4B M4 Pro sweep reproduces that threshold: quality is budget-invariant, fitting all eight

resources eliminates reloads, and mean resolution time falls by about 68%. On a separate 48 GiB M4 Pro, a synchronized Qwen3 Q4 ladder at 4B, 8B, 14B, and 32B evaluates 120 natural-QA model-example pairs. Concatenated native K/V preserves all 120 selected-text sequences. Its warm E2/E0 request ratio approaches parity with scale (1.054, 1.031, 1.018, and 1.011), while its cold/cold ratio rises from 0.932 to 0.976: the cold advantage shrinks rather than grows. Full segmented overhead falls from 1.165 to 1.047. No tested 7/8, 3/4, 2/3, or 1/2 consumer suffix passes the preregistered quality gate at any size, so model scale does not yet justify a reduced-layer BALANCED profile.

# 1 Qualification Snapshot

The engine-specific question is whether selected native K/V can improve the quality-memory-latency frontier in Apple unified memory beyond selected-text E0. The live native executor is validated at E2 and preserves 2,086/2,086 matched E0 outputs while removing 90.4–93.0% of visible prompt tokens. The current unfused path remains slower: macro E2/E0 cost ratios are 1.043 cold, 1.139 warm, 1.133 multi-query, and 1.072 concurrent. E0 is therefore the default when visible selected text is acceptable; E2 is a functional native mechanism whose memory and non-disclosure properties require workload-specific valuation.

|                   |                                                                                                          |
|-------------------|----------------------------------------------------------------------------------------------------------|
| Integration level | E2 validated in-process; no E3 scheduler claim                                                           |
| Model/hardware    | Qwen, Llama, and Gemma-family checks on Apple M5; Qwen3 1.7B–32B and 30B-A3B on M4 Pro                   |
| Workload          | QASPER, HotpotQA, and 2WikiMultiHopQA; frozen selections                                                 |
| Quality           | 2,086/2,086 matched E0/E2 pairs; six additional oracle-transport cohorts preserve aggregate F1 exactly   |
| Context reduction | 90.4–93.0% fewer visible prompt tokens                                                                   |
| TTFT/ITL tails    | Online gateway measured separately; common matched tails <b>NOT MEASURED</b>                             |
| Throughput        | Concurrent E2 inverse-cost ratio 1.072, or 93.3% of E0 rate                                              |
| Model scaling     | Warm concat E2/E0 approaches parity from 1.054 (4B) to 1.011 (32B); cold ratio rises from 0.932 to 0.976 |
| Peak memory       | Unified-memory components and controlled pressure measured; full live frontier open                      |
| Reuse             | HOT/WARM lifecycle and restart recovery measured                                                         |
| Main limitation   | Fuse segmented attention; no reduced consumer-layer profile passes the natural-QA smoke gate             |

**PRA primer.** PRA gives each durable semantic resource stable identity, version, authorization, and selectable intervals. E0 renders one frozen selection as visible text. E1 preserves logical identity while executing an ordinary model path. E2 consumes the identical selection as detached native K/V with correct positions, masks, cache topology, one normalization, cleanup, and isolation. E3 adds a scheduler that owns placement, promotion, eviction, sharing, and batching. MLX prompt or rotating caches retain sequential local state; PRA holds independently selected archive state.

**Deployment recommendation.** **Best validated deployment:** E0 selected text for latency-sensitive serving; E2 for bounded native residency, non-disclosure, or persistent resource reuse after profiling. **Use when:** a selected archive should remain outside the visible prompt. **Do not use when:** a large speed advantage is assumed from model scale alone. **Main remaining gate:** a fused segmented kernel plus a joint local-window/PRA-budget/consumer-layer frontier.

# 2 Introduction

MLX provides an efficient array framework and model runtime for Apple Silicon [1]. MLX-LM adds explicit prompt caches, rotating K/V, optional cache quantization, batching, and an OpenAI-like server. These tools make sequential context reuse practical in unified memory. PRA addresses a different object:

large retained memory → query-selected region → active native detail.

Selected Context session deduplication is owned by the shared PRA runtime; engine-native prefix caching is measured independently. MLX prompt, rotating, and split caches report physical sequential reuse separately from logical visibility and native PRA memory. Unified memory may reduce copy overhead, but it does not make every retained token free to index, keep, or attend to.

This paper studies six questions. First, can selected context coexist with MLX’s exact prompt cache? Second, can a bounded rotating cache plus a selected archive record recover old evidence? Third, what software boundary is needed before an MLX adapter may claim native PRA K/V rather than selected visible text? Fourth, how much native K/V can consumer-layer profiles remove without losing the controlled answer? Fifth, do persistence, request sharing, and natural-text causal controls, original benchmark questions, and quantized residency preserve the mechanism? Sixth, when the runtime itself selects evidence, can we separate routing error from native-K/V consumption error?

### 3 Cache Taxonomy

We keep three mechanisms distinct:

**Prompt cache**

exact reusable prompt-prefix K/V, retained by MLX-LM.

**Rotating cache**

bounded recent sequential K/V, with old positions evicted as the prompt advances.

**PRA memory**

query-addressed non-prefix resources whose address state, native detail, and active selection have separate lifecycles.

Prompt-cache bytes are physical retained state. PRA backing may be much larger than active detail. Rotating capacity bounds direct context but does not create an address for evicted evidence.

### 4 Implementation

The engine-neutral namespace in `src/prax_engine_memory.py` identifies a block by tenant/session scope, resource/version, token interval, consumer layer, immutable model revision, dtype/layout, materialization profile, and position policy. Its residency state machine records indexed-only, off-device, prefetching, resident, pinned, evictable, and invalid states. In MLX, the tier name denotes array residency or backing policy within unified memory; we do not mislabel it as CUDA host-to-device transfer.

`src/prax_adapter.py` exposes the HTTP server as E0. It advertises automatic prefix caching and text fallback, but no native K/V. An in-process `MLXNativeExecutor` is required before the adapter advertises E2, selected interval materialization, logical resource deltas, or explicit prompt cache handles. `src/prax_native.py` now provides that executor.

The design choice is important. Precomputing a prompt cache from selected text still makes that text a sequential prefix. A native executor must instead preserve the selected region’s identity and positional policy while combining its K/V with local K/V under correct model attention.

The implementation encodes immutable resource text once and retains post-RoPE K/V in  $[B, H_{kv}, T, D]$  form for every layer. A request-local cache returns *selected memory plus local sequential K/V* to each attention module and constructs a mask in which every query sees selected memory while local keys remain causal. Query positions advance from the source extent; selected text never appears in the visible prompt. The residency manager deduplicates materialization, pins blocks for request lifetime, and evicts only unpinned detail under a bounded byte budget.

For compact residency, each K and V head receives a symmetric int8 scale over its token and head-width axes. Compact arrays remain the retained object; selection dequantizes them to the model-native dtype before constructing the request-local attention cache. This version measures storage economy, not an int8 attention kernel: active attention still consumes full-precision K/V.

Consumer-layer profiles preserve the source-relative query position even on layers that omit selected detail. Such a layer receives only request-local K/V, but its RoPE offset still advances from the source extent; otherwise a profile sweep would confound memory removal with a positional-frame change. Persisted arrays carry a strict sidecar fingerprint over model and tokenizer revisions, dtype, position policy, consumer profile, and resource version. A mismatch is rejected before arrays enter attention.

Table 1: MLX-LM server smoke. TTFT is milliseconds; warm cached is the mean over repeats two through five.

| Condition         | Exact | Prompt | Cold TTFT | Warm TTFT | Warm cached |
|-------------------|-------|--------|-----------|-----------|-------------|
| None              | 0%    | 24     | 523.6     | 505.2     | 23          |
| Prefix            | 0%    | 333    | 446.4     | 355.2     | 332         |
| Selected          | 100%  | 56     | 415.9     | 355.0     | 55          |
| Prefix + selected | 100%  | 365    | 432.0     | 352.3     | 364         |
| Full context      | 100%  | 1577   | 565.4     | 351.8     | 1576        |

## 5 Environment

For cards and cross-engine qualification, we project measurements into *No PRA*, *PRA-No Adaptor*, and *PRA-Adaptor Bundle*. E0 and E2 remain internal mechanism labels: selected-text E0 is the original-model execution and native E2 is PRA without learned adaptor behavior. The bundle condition requires a recorded immutable bundle revision and is otherwise `NOT_MEASURED`. Both PRA deltas use No PRA as baseline, preserve sign, and declare metric unit, aggregation, and direction explicitly. The host was an Apple M5 MacBook Pro with 10 CPU and 10 GPU cores, 16 GB unified memory, Metal 4, and macOS 26.4.1. The environment used Python 3.12.7, MLX 0.32.0, and mlx-lm 0.31.3. The model was `mlx-community/Qwen3-0.6B-4bit` at revision `73e3e38d981303bc594367cd910ea6eb48349da8`. The HTTP server retained up to 16 prompt-cache entries. It emits an explicit warning that its basic security checks are not production-grade; we treat it as an experimental deployment baseline.

The end-task study uses `mlx-community/Qwen3-1.7B-4bit`, MLX-LM 0.31.3, five seeds, and four held-out examples per seed and dataset. Evidence is oracle-scoped from dataset annotations and truncated to 384 tokens; routing quality is deliberately outside the first engine-transport experiment. A replication uses an M4 Pro with 20 GPU cores and 48 GiB unified memory, MLX 0.32.0, and MLX-LM 0.31.3. It repeats the same five seeds for Qwen3-1.7B and Qwen3-4B. Thus hardware and model scale change while dataset selection and execution conditions remain fixed. A second matched-budget cohort retains every HotpotQA and 2Wiki candidate document and every QASPER paper paragraph, then routes four candidates before native materialization.

## 6 Server Prompt-Cache Experiment

The shared fixed task contains a stable 309-token prefix, one authoritative codeword record, twelve distractor records, and a short query. Five conditions separate no memory, prefix only, selected memory only, prefix plus selected memory, and full context. Each is repeated five times. The server reports cached tokens in the streaming usage object, and SSE events determine TTFT.

Table 1 shows correct controls: selected and full conditions are 5/5, while conditions without evidence are 0/5. Prefix plus selected memory uses 365 tokens and full context uses 1,577, again a 76.9% visible reduction. Warm selected requests reuse 364 tokens; warm full requests reuse 1,576. The server’s nearest-prefix cache therefore composes cleanly with the selected-text baseline.

Warm TTFT is nearly flat, 352.3 versus 351.8 ms, because exact prompt caches remove most repeated prefix work in both conditions. This is a useful negative latency result: after a perfect repeat cache hit, reducing visible input does not improve warm TTFT on this setup. Physical retention still differs. The log shows five distinct prompt-cache sequences occupying about 0.28 GB at the end of the matrix.

## 7 Rotating Cache and Selected Archive

The Python-API control uses five deterministic filler permutations. One authoritative four-digit fact appears near the beginning of a roughly thousand-token prompt. We compare full sequential K/V with rotating capacities 64, 128, 256, and 512. For each rotating capacity, a second condition reintroduces the authoritative record immediately before the final question. The answer metric accepts harmless formatting but requires the correct digits.

Full sequential K/V is 5/5 with 112 MB of cache. Rotating-only is 0/5 at every capacity while reducing layer-cache storage from 112 MB to 7, 14, 28, and 56 MB. Selected archive text reaches 2/5 at 64 and 0/5 at larger capacities. Mean completion latency is 469 ms for full K/V and approximately 705–736 ms for bounded conditions. The generated outputs show chat-format and answer-path instability, not merely loss of the early fact.

Table 2: Five-seed controlled rotating-cache experiment. Cache MB is the sum of layer-cache `nbytes`; latency is the mean completion time.

| Condition                   | K/V tokens | Exact | Cache MB | Latency ms |
|-----------------------------|------------|-------|----------|------------|
| Full sequential K/V         | full       | 100%  | 112.0    | 468.9      |
| Rotating only               | 64         | 0%    | 7.0      | 706.6      |
| Rotating only               | 128        | 0%    | 14.0     | 730.2      |
| Rotating only               | 256        | 0%    | 28.0     | 722.1      |
| Rotating only               | 512        | 0%    | 56.0     | 721.2      |
| Rotating + selected archive | 64         | 40%   | 7.0      | 704.7      |
| Rotating + selected archive | 128        | 0%    | 14.0     | 736.2      |
| Rotating + selected archive | 256        | 0%    | 28.0     | 713.1      |
| Rotating + selected archive | 512        | 0%    | 56.0     | 732.2      |

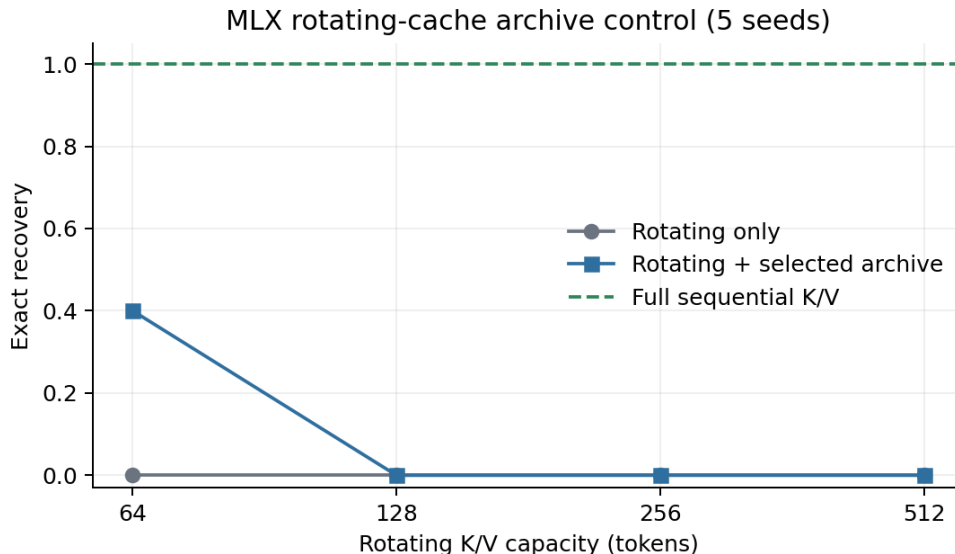


Figure 1: A bounded rotating cache is not a reliable archive in this control. Selected late text recovers two of five cases only at the smallest capacity.

This non-monotonic result should not be read as evidence that smaller cache is better. It indicates that the direct rotating-cache intervention changes more than evidence availability for this model and prompt. The selected-text fallback does not restore stable computation. A native PRA executor must retain local sequential semantics while adding selected memory separately; the next section evaluates that mechanism directly.

Peak process memory remains about 1.01–1.07 GB across these conditions despite the 7–112 MB cache range. Model and runtime allocations dominate on this small checkpoint, so layer-cache bytes are the interpretable economy metric. Larger models and memory pressure are required before process-level savings claims.

## 8 Native Selected-K/V Experiment

The source contains 141 tokens. Ordinary split-cache and native selected-K/V produce identical logits, identical argmax tokens, and 5/5 exact recovery. Native selected execution averages 166.3 ms after a separately measured 55.4 ms one-time encoding step and retains 15.4 MB of all-layer K/V. Most importantly, replacing the local sequential cache by a 64-token rotating cache still recovers 5/5. The selected archive is therefore independent of local rotation rather than late visible text competing inside that bounded cache.

Table 4 repeats the mechanism on Llama 3.2 1B and Gemma 3 1B. Llama achieves 5/5 native and rotating-local recovery with exact split-cache logits while retaining 4.34 MB. Gemma also has zero logit error and matching argmax decisions, but both its ordinary and native conditions are 0/5 on this particular answer-format task. Gemma

Table 3: Five-seed native selected-K/V control. Latency excludes the separately reported one-time native encoding cost. Native MB is all-layer selected K/V.

| Condition                  | Exact | Latency ms | Native MB | Max error |
|----------------------------|-------|------------|-----------|-----------|
| Ordinary split cache       | 100%  | 187.0      | 0.0       | 0         |
| Native selected K/V        | 100%  | 166.3      | 15.4      | 0         |
| Native + rotating local 64 | 100%  | 215.5      | 15.4      | 0         |

Table 4: Cross-model native selected-K/V replication. Native and rotating columns report exact recovery over five seeds; error is zero for every row.

| Model              | Seeds | Native exact | Rotating exact | Native MB |
|--------------------|-------|--------------|----------------|-----------|
| Qwen3 0.6B 4-bit   | 5     | 100%         | 100%           | 15.42     |
| Llama 3.2 1B 4-bit | 5     | 100%         | 100%           | 4.34      |
| Gemma 3 1B 4-bit   | 5     | 0%           | 0%             | 3.58      |

therefore establishes mechanism parity, not answer recovery. Together the runs cover three attention topologies and tokenizers; they are not yet natural-workload results.

## 9 Profiles, Persistence, and Concurrency

Table 5: Five-seed consumer-layer sweep. Active MB counts selected native K/V, not model weights or request-local K/V.

| Consumer profile | Layers | Exact | Active MB | Max error |
|------------------|--------|-------|-----------|-----------|
| all              | 28     | 100%  | 15.42     | 0         |
| disabled         | 0      | 0%    | 0.00      | 6.188     |
| last_4           | 4      | 0%    | 2.20      | 4.844     |
| last_half        | 14     | 100%  | 7.71      | 4.031     |
| last_quarter     | 7      | 0%    | 3.86      | 4.188     |

Table 5 exposes a sharp controlled boundary. Full native memory and the last-half profile both recover 5/5 answers. Last quarter, last four, and disabled profiles recover 0/5. Last half reduces active selected K/V from 15.42 to 7.71 MiB. Its nonzero logit difference is expected because fourteen layers no longer see the archive; exact recovery, not bit parity, is the profile criterion. This is a candidate economy profile for this fixture, not a production profile calibrated from five examples.

Table 6: Five-seed persistence and concurrent request sharing. Persist load is the mean for a 15.43 MiB selected-K/V sidecar.

| Concurrency | Exact | Requests/s | Mean latency ms | Shared MB saved | Persist load ms |
|-------------|-------|------------|-----------------|-----------------|-----------------|
| 1           | 100%  | 6.04       | 164.9           | 0.0             | 7.2             |
| 2           | 100%  | 10.80      | 180.1           | 15.4            | 7.2             |
| 4           | 100%  | 16.98      | 222.4           | 46.3            | 7.2             |
| 8           | 100%  | 16.01      | 383.9           | 108.0           | 7.2             |

Persisting and reloading native arrays yields zero maximum logit error. Mean save and load times are 28.7 and 7.2 ms. Separate request-local caches share one immutable memory object: at concurrency eight this avoids 108.0 MiB of duplicate selected K/V. Aggregate throughput increases through concurrency four and then plateaus, while mean per-request latency rises from 164.9 to 383.9 ms. These are in-process controlled measurements, not queued server tail latency.

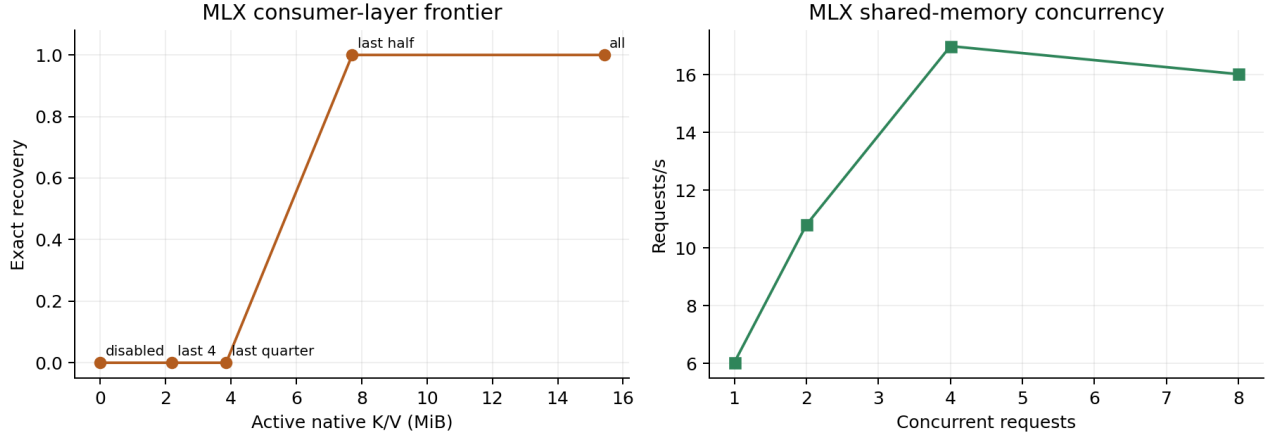


Figure 2: Left: the controlled consumer-layer quality–memory boundary. Right: shared immutable K/V raises throughput until the small Apple host saturates.

## 10 Larger-Model Segmented-Attention Calibration

The 48 GiB M4 Pro permits a single-process Q4 ladder that was unavailable to the original small-host study. We load pinned Qwen3-8B, 14B, and 32B checkpoints and Qwen3-30B-A3B. Each model evaluates five unique annotated evidence examples from each of QASPER, HotpotQA, and 2Wiki. The selected source is bounded at 384 tokens and held fixed across selected-text E0, concatenated native E2, live segmented native E2, and six contiguous consumer profiles. Retrieval is absent from this experiment. It asks whether MLX transports the same selected evidence and how sensitivity to consumer depth scales.

Table 7: M4 Pro larger-model calibration over 15 natural-QA examples per model. Agreement is greedy sequence agreement with selected-text E0;  $|\Delta|LP$  is the mean across dataset-level absolute gold-log-probability deltas. PRA MiB is active all-layer selected K/V, excluding weights and local K/V.

| Model         | Layers | E0 F1 | Concat agr. | Seg. agr. | $ \Delta LP$ | PRA MiB | Seg./E0 |
|---------------|--------|-------|-------------|-----------|--------------|---------|---------|
| Qwen3-8B      | 36     | 0.236 | 1.000       | 1.000     | 0.189        | 39.6    | 1.08    |
| Qwen3-14B     | 40     | 0.277 | 1.000       | 1.000     | 0.027        | 43.9    | 1.05    |
| Qwen3-32B     | 64     | 0.231 | 1.000       | 0.867     | 0.039        | 70.3    | 1.03    |
| Qwen3-30B-A3B | 48     | 0.256 | 1.000       | 0.933     | 0.146        | 26.4    | 1.12    |

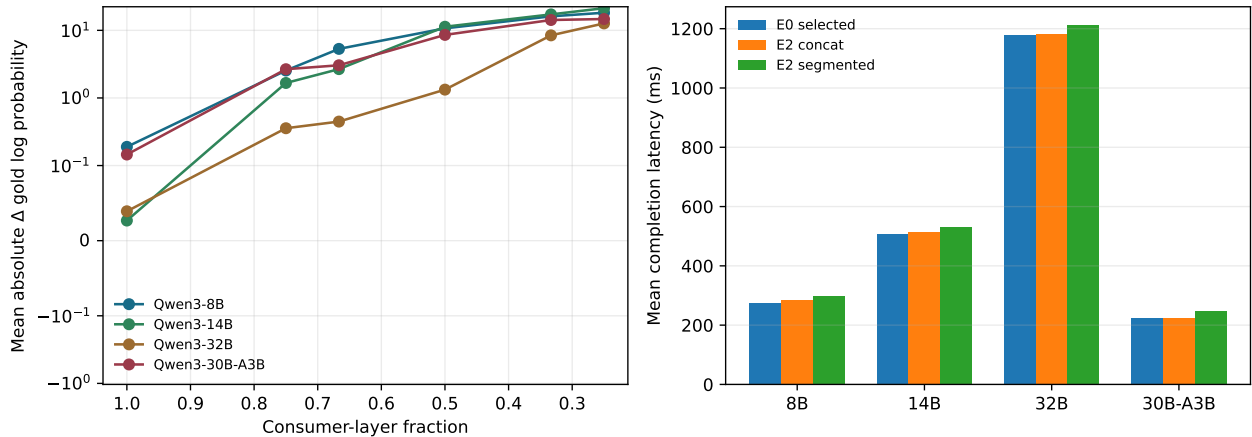


Figure 3: Left: likelihood sensitivity as contiguous late consumers are removed. Dense 32B is less sensitive in likelihood, but reduced profiles still change generated sequences. Right: selected text, concatenated native K/V, and the unfused segmented implementation on the same selected evidence.

Concatenated E2 is the numerical reference path: all 60 model-example pairs match E0 sequences. The segmented path performs the same mathematical normalization with separate memory and local score blocks, accumulating the shared denominator and value numerator in FP32. It matches every 8B and 14B sequence and 27/30 across dense 32B and MoE. The remaining changes are tie-sensitive numerical differences after 48–64 repeated attention layers, not evidence-selection differences; mean absolute answer-likelihood drift stays at or below .189 nats. The current Python/MLX composition is also not a speed optimization: it costs 1.08, 1.05, 1.03, and 1.12 times E0 completion latency. Its systems result is removal of the temporary K/V concatenation, not a fused-kernel throughput claim.

The placement result is more conditional. Across the dense ladder, mean absolute likelihood loss for the last-three-quarter profile falls from 2.545 to 1.665 to .356 nats. This supports the hypothesis that larger dense models can be less likelihood-sensitive to omitted early consumers. Yet sequence agreement for that profile is only .267, .467, and .133, respectively. The 30B-A3B point is fast and uses 26.4 MiB of active all-layer K/V versus 70.3 MiB for dense 32B, but its last-three-quarter loss is 2.662 nats. Parameter sparsity therefore does not imply semantic-context sparsity. All reduced-layer rows remain CALIBRATION PENDING; the correctness profile remains all eligible layers.

## 10.1 Matched lifecycle scaling law

The preceding profile experiment synchronized ordinary source encoding, but an earlier 1.7B/4B runner did not. MLX executes lazily: that runner called the ordinary source forward outside its timer without calling `mx.eval`, so the work completed inside generation. Native encoding did call `mx.eval`. Its apparent 44.5–54.5% E2/E0 completion ratio therefore mixed cold E0 with warm E2 and is retired.

The corrected runner makes lifecycle state part of the metric. *Cold* cost is synchronized representation encoding plus query generation; *warm* cost is generation from synchronized reusable ordinary or native state. It evaluates two independently shuffled examples per dataset under each of five seeds, or 30 examples per model, at 384 selected source tokens and 12 greedy output tokens. The 4B anchor and pinned 8B, 14B, and 32B models each run ordinary warm E0, concatenated warm E2, and segmented E2 at all, 7/8, 3/4, 2/3, and 1/2 of decoder layers. This produces 840 model-backed condition rows; retrieval and selected evidence are fixed within each pair.



Table 8: Matched lifecycle scaling on the M4 Pro. Ratios divide E2 by E0; lower is faster. Concat is the numerical reference. Segmented agreement is exact greedy-sequence agreement. “None” means no reduced profile passed the smoke quality gate.

| Model     | $n$ | concat warm | concat cold | seg. warm | seg. agr. | balanced | PRA MiB |
|-----------|-----|-------------|-------------|-----------|-----------|----------|---------|
| Qwen3-4B  | 30  | 1.054       | 0.932       | 1.165     | 0.833     | none     | 52.6    |
| Qwen3-8B  | 30  | 1.031       | 0.932       | 1.102     | 0.767     | none     | 52.6    |
| Qwen3-14B | 30  | 1.018       | 0.950       | 1.055     | 0.933     | none     | 58.5    |
| Qwen3-32B | 30  | 1.011       | 0.976       | 1.047     | 0.833     | none     | 93.6    |

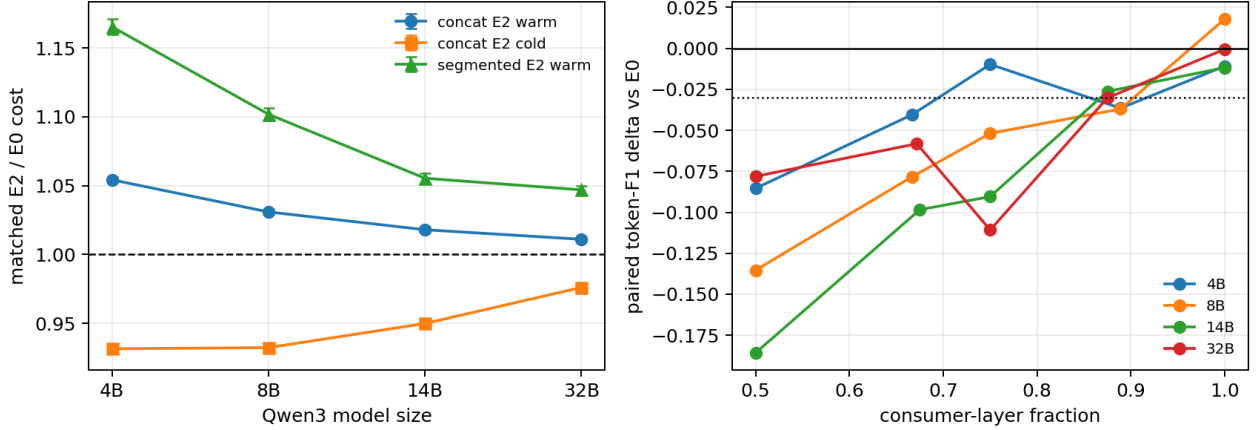


Figure 4: Left: matched cold and warm E2/E0 costs. Warm overhead approaches parity with model size, but cold E2’s advantage contracts. Right: paired task-F1 change as consumer layers are removed; the dotted line is the  $-0.03$  smoke margin, not a production noninferiority bound.

Concatenated E2 preserves every one of the 120 E0 sequences and has zero gold-answer-likelihood drift. Its warm ratio falls monotonically from 1.054 at 4B to 1.011 at 32B. Paired bootstrap 95% intervals are  $[1.052, 1.056]$  and  $[1.009, 1.013]$  at the endpoints. This four-point trend is convergence toward parity rather than a native execution advantage. The cold ratio moves in the opposite direction, from 0.932 at 4B and 8B to 0.950 at 14B and 0.976 at 32B. Reference encoding and ordinary source-cache construction scale similarly, so larger models do not amplify the measured cold benefit.

The unfused segmented path also improves with scale: all-layer warm overhead falls from 1.165 to 1.047. Its exact agreement is 0.833, 0.767, 0.933, and 0.833, while mean absolute gold-log-probability drift remains 0.090–0.214 nats. These are small reduction-order effects accumulated across many layers; the concatenated reference shows that source  $K/V$  itself is exact.

Table 9: Consumer-depth quality and cost.  $|\Delta|LP$  is mean absolute gold-answer-log-probability change from E0. Agreement is exact sequence agreement. Reduced profiles save active PRA K/V proportionally but none passes both the paired F1 and likelihood gate.

| Model     | profile  | fraction | $\Delta F1$ | $ \Delta LP$ | agreement | warm ratio |
|-----------|----------|----------|-------------|--------------|-----------|------------|
| Qwen3-4B  | all      | 1.000    | -0.011      | 0.214        | 0.833     | 1.165      |
| Qwen3-4B  | last 7/8 | 0.889    | -0.037      | 2.131        | 0.300     | 1.177      |
| Qwen3-4B  | last 3/4 | 0.750    | -0.010      | 3.412        | 0.233     | 1.155      |
| Qwen3-4B  | last 2/3 | 0.667    | -0.040      | 7.029        | 0.167     | 1.140      |
| Qwen3-4B  | last 1/2 | 0.500    | -0.085      | 14.086       | 0.167     | 1.115      |
| Qwen3-8B  | all      | 1.000    | +0.018      | 0.167        | 0.767     | 1.102      |
| Qwen3-8B  | last 7/8 | 0.889    | -0.037      | 1.758        | 0.267     | 1.092      |
| Qwen3-8B  | last 3/4 | 0.750    | -0.052      | 3.825        | 0.200     | 1.078      |
| Qwen3-8B  | last 2/3 | 0.667    | -0.078      | 7.019        | 0.233     | 1.070      |
| Qwen3-8B  | last 1/2 | 0.500    | -0.135      | 12.587       | 0.133     | 1.053      |
| Qwen3-14B | all      | 1.000    | -0.012      | 0.114        | 0.933     | 1.055      |
| Qwen3-14B | last 7/8 | 0.875    | -0.026      | 1.298        | 0.333     | 1.053      |
| Qwen3-14B | last 3/4 | 0.750    | -0.090      | 2.148        | 0.333     | 1.043      |
| Qwen3-14B | last 2/3 | 0.675    | -0.098      | 2.719        | 0.333     | 1.035      |
| Qwen3-14B | last 1/2 | 0.500    | -0.186      | 11.460       | 0.133     | 1.023      |
| Qwen3-32B | all      | 1.000    | -0.001      | 0.090        | 0.833     | 1.047      |
| Qwen3-32B | last 7/8 | 0.875    | -0.030      | 1.009        | 0.167     | 1.032      |
| Qwen3-32B | last 3/4 | 0.750    | -0.111      | 1.495        | 0.167     | 1.025      |
| Qwen3-32B | last 2/3 | 0.672    | -0.058      | 1.791        | 0.100     | 1.020      |
| Qwen3-32B | last 1/2 | 0.500    | -0.078      | 2.173        | 0.133     | 1.011      |

The consumer-layer hypothesis is not supported under this evidence geometry. At 32B, the last-7/8 profile loses 0.030 mean F1, has a wide paired bootstrap interval  $[-0.130, 0.058]$ , and drifts by 1.009 nats in absolute answer likelihood. The last-half profile reaches near-E0 warm cost (1.011) but loses 0.078 F1 and drifts by 2.173 nats. Similar or larger likelihood losses occur at smaller sizes. The minimum supported consumer fraction therefore remains one. The SDK’s `BALANCED` profile consequently consumes native memory at all eligible layers. Reduced profiles remain `CALIBRATION PENDING`; parameter scale alone does not promote them.

**Public exact-identity qualification.** The release pipeline imports these measurements into model cards only after matching the model ID, immutable revision, 4-bit MLX quantization, `mlx-lm` version, `BALANCED` profile, and Native Memory execution mode. The public Qwen3-8B, 14B, and 32B cards use the concatenated all-consumer path, not the numerically drifting segmented variant. Across the 15 paired QASPER, HotpotQA, and 2Wiki examples at each size, token F1 is unchanged and all outputs agree exactly; visible prompt tokens fall by 89.1%. Mean completion overhead contracts from 3.0% to 1.1% and 0.5% with scale. The 32B median TTFT point estimate is 6.6% lower, but the cohort is too small for a tail- latency claim. We therefore label these rows `ENGINE QUALIFIED`, retain all eligible consumer layers in `BALANCED`, and leave `QUALITY` and `ECONOMY` as `CALIBRATION PENDING`.

## 10.2 M5 timing replication and held-out layer gate

We repeated the corrected runner on a 16 GiB M5 after one unmeasured graph warm-up per execution path. Each measured checkpoint uses two examples from each dataset under five seeds, or 30 paired examples per model. Qwen3-4B, 8B, and 14B complete; the 32B preflight requires 22.86 GB against 17.18 GB physical memory and is recorded as `NOT_RUN_CAPACITY_GATE`.

Table 10: Corrected M5 replication. Ratios compare matched warm or cold states; concat preserves every E0 output. No reduced profile passes the quality gate.

| Model     | $n$ | concat warm | concat cold | seg. warm | seg. agr. | balanced | PRA MiB |
|-----------|-----|-------------|-------------|-----------|-----------|----------|---------|
| Qwen3-4B  | 30  | 1.035       | 0.973       | 1.290     | 0.933     | none     | 52.6    |
| Qwen3-8B  | 30  | 1.015       | 0.968       | 1.156     | 0.867     | none     | 52.6    |
| Qwen3-14B | 30  | 0.980       | 0.946       | 1.031     | 0.933     | none     | 58.5    |

Warm concat ratios fall from 1.035 at 4B to 1.015 at 8B and 0.980 at 14B. The 14B paired-bootstrap interval,  $[0.922, 1.012]$ , includes parity, so the point estimate is promising rather than a confirmed native speedup. Cold usable-context ratios are 0.973, 0.968, and 0.946. Unfused segmented overhead also contracts from 1.290 to 1.156 and 1.031; agreement remains 0.933, 0.867, and 0.933. The M5 result therefore strengthens model-scale convergence and the exact concat reference. It does not justify a fused-kernel claim or a reduced consumer profile.

We also calibrated a parameter-free noncontiguous layer selector on Qwen3-4B. Seed 11 greedily tests eight contiguous layer groups under a 0.05-nat absolute-likelihood bound and 0.90 first-token agreement, then seeds 23/37/53/71 form the held-out cohort. Every candidate removal fails on the calibration split; even dropping the first five layers changes likelihood by 1.538 nats. The selected set is therefore all 36 layers.

Table 11: Held-out noncontiguous consumer-layer calibration. The calibrated gate is learned from the calibration split but resolves to all layers.

| Condition         | $n$ | Layers | F1    | Sequence agreement | Completion ms |
|-------------------|-----|--------|-------|--------------------|---------------|
| Selected text E0  | 12  | 0/36   | 0.224 | 1.000              | 356.1         |
| Segmented E2, all | 12  | 36/36  | 0.224 | 1.000              | 469.3         |
| Calibrated gate   | 12  | 36/36  | 0.224 | 1.000              | 470.5         |

The 12 held-out examples preserve every selected-text sequence, but because the calibrated action equals the all-layer action this is a negative sparsity result. A trained, query-conditioned gate remains open work. Fixed suffixes and calibration-only group elimination have both failed.

### 10.3 Cross-family replication at 8B

The concat transport result is not Qwen-specific. On a 48 GiB M4 Pro we run Llama 3.1-8B-Instruct-4bit on 60 natural-QA examples: 20 each from QASPER, HotpotQA, and 2Wiki, with five seeds. The selector output, source cap, and 12-token greedy decode are frozen between E0 and E2. Native E2 preserves all 60 output sequences, aggregate token F1 (0.3563), and gold-answer log probability ( $-6.0414$ ). Warm request cost is 349.4 versus 338.7 ms, a 1.032 ratio; cold time to usable context is 1,076.1 versus 1,151.2 ms, a 0.935 ratio. Mean active native detail is 46.2 MiB. A current-code 45-example overlapping rerun reproduces exact output and likelihood parity with warm/cold ratios 1.032/0.939. This closes useful-scale transport replication for a second model family, but not reduced-layer calibration: the segmented consumer patch remains Qwen-specific.

Table 12: Llama 3.1 8B cross-family transport replication over 60 natural-QA examples. Times are means in milliseconds.

| Mode             | F1     | Gold log-prob. | Warm  | Cold usable | Agreement |
|------------------|--------|----------------|-------|-------------|-----------|
| E0 selected text | 0.3563 | -6.0414        | 338.7 | 1,151.2     | 60/60     |
| E2 native concat | 0.3563 | -6.0414        | 349.4 | 1,076.1     | 60/60     |

## 11 Long-Context Segmented Execution

The preceding ladder fixes selected evidence at a short context. We next keep the same 384-token selection and add deterministic same-dataset distractors to 8K source tokens for Qwen3-8B, 14B, and 32B. Five unique examples per dataset give 15 independent questions per model. An 8B boundary run adds one example per dataset at 32K. The requested 64K rows are emitted as NOT\_RUN\_MODEL\_CONTEXT\_LIMIT: the pinned models expose a 40,960 position limit, and the harness does not silently truncate.

Four representations separate execution questions. FULL VISIBLE is the ordinary prefix. E0 SELECTED renders the frozen evidence as text. E2 SELECTED consumes that selection through segmented native K/V. E2 SOURCE RELATIVE consumes the full dense source through the same segmented path; QUERY RESTART reuses its K/V but incorrectly starts the query at position zero.

Table 13: Long-context MLX calibration.  $\Delta LP$  is full visible minus selected text. Selected E2 agreement compares the two frozen selected-evidence representations. Source/restart columns compare full native K/V with the ordinary full prefix.

| Model     | Context | $\Delta LP$ full-selected | selected E2 agr. | source RMSE | restart RMSE | source agr. | restart agr. |
|-----------|---------|---------------------------|------------------|-------------|--------------|-------------|--------------|
| Qwen3-8B  | 8K      | +1.192                    | 1.000            | 0.112       | 2.572        | 0.667       | 0.000        |
| Qwen3-8B  | 32K     | -0.603                    | 1.000            | 0.128       | 2.530        | 0.333       | 0.000        |
| Qwen3-14B | 8K      | -1.485                    | 1.000            | 0.100       | 4.786        | 0.933       | 0.000        |
| Qwen3-32B | 8K      | -0.999                    | 0.867            | 0.121       | 3.671        | 0.667       | 0.000        |

Selected E2 preserves every selected-text sequence at 8B and 14B. At 32B it preserves 13/15 sequences; mean gold-answer log probability differs by only +0.030 nats and mean F1 by +0.017. The full-source segmented path is deliberately held to a stricter ordinary-prefix comparison: first-logit RMSE is 0.112 and 0.100 at 8K, whereas query restart gives 2.572 and 4.786. Source-relative sequence agreement is 0.667 and 0.933; restart agreement is zero. The segmented implementation uses one mathematical normalization but separate FP32 reductions, so repeated unfused numerical differences accumulate through depth. This is distinct from the concatenated-cache reference above, which preserves all 60 sequences.

The pressure result illustrates why full-source native K/V is not the product goal. At 8B, selected detail occupies 39.6 MiB at both 8K and 32K, while all-layer full-source K/V grows from 1,152 to 4,608 MiB. MLX allocator peak for the ordinary full condition rises from 9.10 to 22.55 GiB. At 14B and 8K, selected detail is 43.9 MiB versus 1,280 MiB full-source K/V. At 32B the corresponding pair is 70.3 versus 2,048 MiB, while ordinary full-prefix peak allocation reaches 25.62 GiB. The three-case 8B 32K boundary is useful for mechanism and pressure, not tail-latency inference.

**Corrected occupied-context curve.** An independent Qwen3-8B rerun uses six natural-QA examples at each of 2K, 8K, and 32K occupied source tokens. Selection remains fixed at the same bounded evidence, and each execution path is warmed before measurement.

Table 14: Corrected occupied-context pressure. Selected detail stays fixed as the distractor-bearing visible context grows. Times are in-process completion latencies and do not estimate queued tails.

| Context | $n$ | E0 selected ms | E2 selected ms | E2/E0 | Selected MiB | Full peak GiB |
|---------|-----|----------------|----------------|-------|--------------|---------------|
| 2K      | 6   | 350.9          | 387.4          | 1.104 | 39.6         | 6.37          |
| 8K      | 6   | 350.9          | 387.4          | 1.104 | 39.6         | 11.44         |
| 32K     | 6   | 350.9          | 387.4          | 1.104 | 39.6         | 31.85         |

Selected E2 agrees with selected-text E0 on all 18 sequences and keeps active detail at 39.6 MiB. Its mean completion time is 387.4 ms at every occupancy, 1.104 times the 350.9 ms E0 selected path. In contrast, full-visible peak allocation grows from 6.37 to 31.85 GiB and completion time from 387.1 to 1,116.0 ms. Full-source native K/V grows from 288 to 4,608 MiB and is not the intended product mode. Query restart fails on every 8K and 32K sequence, confirming that source-relative position continuity is a correctness requirement rather than optional metadata.

## 12 Natural-Text Causal Controls

The QASPER and HotpotQA builders draw natural paper or article text and place a balanced **AnswerCode** **yes/no** anchor among 63 source units. They test whether an answer-bearing natural-text resource survives the native transport; they do not claim unrestricted question answering. Each dataset contributes eight examples under each of five seeds, for 40 examples per condition. We rank the next-token logits for **yes** and **no**; free-form generation is retained only as a diagnostic because this small completion model often ignores the requested answer format.

Ordinary split-cache and all-layer native execution agree on all 80 examples and have identical mean margins within each dataset. Last-half execution also ranks all 80 correctly while reducing the margin. Disabled memory is

Table 15: Natural-text transport controls. Margin is the mean gold-minus-other answer logit.

| Dataset  | Condition        | Samples | Ranked exact | Margin |
|----------|------------------|---------|--------------|--------|
| qasper   | native_all       | 40      | 100%         | 2.953  |
| qasper   | native_disabled  | 40      | 50%          | 0.014  |
| qasper   | native_last_half | 40      | 100%         | 1.009  |
| qasper   | native_shuffled  | 40      | 50%          | -0.108 |
| qasper   | ordinary_split   | 40      | 100%         | 2.953  |
| hotpotqa | native_all       | 40      | 100%         | 2.885  |
| hotpotqa | native_disabled  | 40      | 52%          | -0.002 |
| hotpotqa | native_last_half | 40      | 100%         | 1.113  |
| hotpotqa | native_shuffled  | 40      | 50%          | 0.166  |
| hotpotqa | ordinary_split   | 40      | 100%         | 2.885  |

50.0% on QASPER and 52.5% on HotpotQA; shuffled memory is 50.0% on both. The causal controls therefore rule out query-only label bias as the source of native success for this transport task.

### 13 Original-Answer QA and Quantized Residency

The next experiment removes the inserted answer code. It evaluates original QASPER, HotpotQA, and 2Wiki-MultiHopQA questions and short gold answers. For each example, annotated evidence is placed first and natural distractor or abstract text fills the 384-token source budget. This is an oracle-evidence materialization study: it asks whether the engine transports useful native K/V, not whether PRA routing found the evidence.

Table 16: Natural-QA generation with Qwen3-1.7B-4bit over five seeds. Strict EM is zero because the model often continues after the answer; token F1 and total gold-answer log-probability retain the useful quality signal.

| Dataset         | Condition            | $n$ | Token F1 | Gold log-prob. | Resident MiB |
|-----------------|----------------------|-----|----------|----------------|--------------|
| qasper          | Ordinary split K/V   | 20  | 0.136    | -28.66         | 0.0          |
| qasper          | Native FP K/V        | 20  | 0.136    | -28.66         | 37.6         |
| qasper          | Native int8 resident | 20  | 0.147    | -27.73         | 18.8         |
| qasper          | Shuffled native K/V  | 20  | 0.057    | -52.18         | 37.6         |
| qasper          | No memory            | 20  | 0.042    | -54.53         | 0.0          |
| hotpotqa        | Ordinary split K/V   | 20  | 0.129    | -10.95         | 0.0          |
| hotpotqa        | Native FP K/V        | 20  | 0.129    | -10.95         | 42.0         |
| hotpotqa        | Native int8 resident | 20  | 0.140    | -10.21         | 21.0         |
| hotpotqa        | Shuffled native K/V  | 20  | 0.045    | -19.48         | 42.0         |
| hotpotqa        | No memory            | 20  | 0.058    | -15.68         | 0.0          |
| 2wikimultihopqa | Ordinary split K/V   | 20  | 0.059    | -10.30         | 0.0          |
| 2wikimultihopqa | Native FP K/V        | 20  | 0.059    | -10.30         | 42.0         |
| 2wikimultihopqa | Native int8 resident | 20  | 0.059    | -9.92          | 21.0         |
| 2wikimultihopqa | Shuffled native K/V  | 20  | 0.060    | -16.39         | 42.0         |
| 2wikimultihopqa | No memory            | 20  | 0.073    | -13.88         | 0.0          |

Table 16 shows exact full-precision transport parity: ordinary split-cache and native K/V have identical aggregate token F1 and gold answer log-probability on every completed dataset. On QASPER the two paths both reach 0.136 F1 and  $-28.66$  mean log-probability; on HotpotQA both reach 0.129 and  $-10.95$ ; on 2Wiki both reach 0.059 and  $-10.30$ . Shuffling memory lowers F1 to 0.057 and 0.045 on QASPER and HotpotQA and makes gold log-probability substantially worse on all three datasets. On 2Wiki, shuffled and no-memory F1 are slightly higher despite worse gold log-probability, exposing verbose generation as a noisy metric rather than supporting a quality gain. The exact transport parity and answer-likelihood controls establish source-dependent computation; broader decoding evaluation remains necessary.

Per-head int8 residency reduces mean selected storage from 37.6 to 18.8 MiB on QASPER and from 42.0 to 21.0 MiB on HotpotQA. Its aggregate F1 and gold log-probability are slightly higher in this small cohort, but this is not evidence that quantization improves quality: generation is discrete and the sample is small. The supported conclusion is non-degradation here plus a near-exact  $2\times$  residency reduction. Dequantization restores model-native K/V before attention, so active materialization bytes are not reduced.

### 13.1 M4 Pro cross-model replication

The M4 Pro replication repeats all five execution conditions for 20 examples per dataset and model. It is independent of the earlier M5 run and adds Qwen3-4B. Table 17 reports the selector-frozen oracle-evidence comparison; Figure 5 keeps transport quality separate from the shuffled-memory causal control.

Table 17: Cross-model original-answer QA on the M4 Pro. E0 is ordinary split-cache execution; E2 is the same selected evidence as native K/V. FP/int8 is retained selected-K/V residency; int8 is dequantized before attention. The original mixed-state latency column is retired.

| Model      | Dataset  | $n$ | E0 F1 | E2 F1 | Shuf. F1 | FP/int8 MiB |
|------------|----------|-----|-------|-------|----------|-------------|
| Qwen3-1.7B | 2Wiki    | 20  | 0.059 | 0.059 | 0.060    | 42.0/21.0   |
| Qwen3-1.7B | HotpotQA | 20  | 0.129 | 0.129 | 0.045    | 42.0/21.0   |
| Qwen3-1.7B | QASPER   | 20  | 0.136 | 0.136 | 0.057    | 37.6/18.8   |
| Qwen3-4B   | 2Wiki    | 20  | 0.097 | 0.097 | 0.011    | 54.0/27.0   |
| Qwen3-4B   | HotpotQA | 20  | 0.212 | 0.212 | 0.028    | 54.0/27.0   |
| Qwen3-4B   | QASPER   | 20  | 0.249 | 0.249 | 0.019    | 48.3/24.2   |

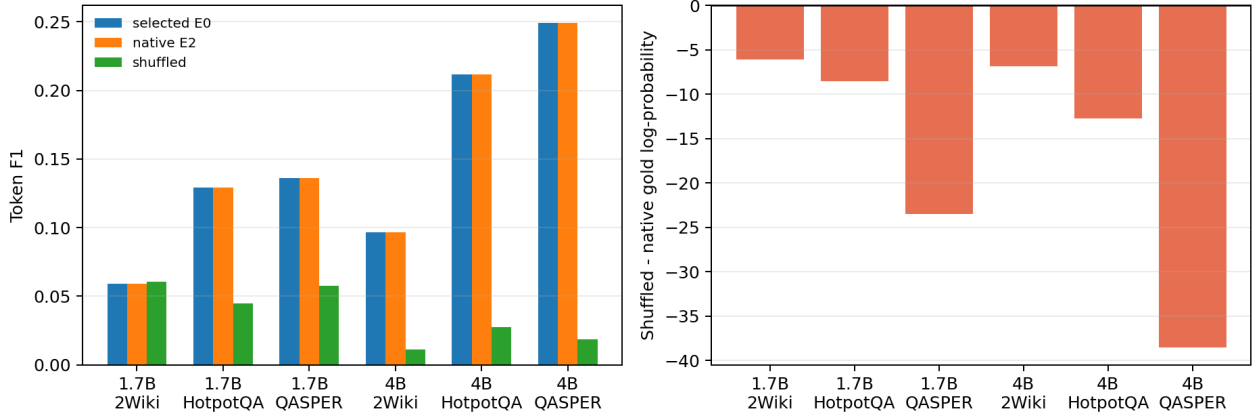


Figure 5: Cross-model transport and causal controls. Full-precision native E2 matches ordinary split-cache aggregate F1 in every cohort, while shuffled memory degrades quality and gold-answer likelihood.

Aggregate E0/E2 F1 is identical in all six cohorts. The 4B model improves E0/E2 F1 from 0.070 to 0.097 on 2Wiki, 0.143 to 0.212 on HotpotQA, and 0.153 to 0.249 on QASPER. Shuffled 4B memory falls to 0.011, 0.028, and 0.019, respectively, while its gold-answer log-probability drops by 6.81–38.58 nats relative to native E2. These interventions support a source-dependent native computation rather than query-only answer bias.

Full-precision residency is 37.6–42.0 MiB at 1.7B and 48.3–54.0 MiB at 4B; compact int8 residency halves each value. This is a retained-storage result: request-local dequantization still restores full-precision active K/V. The former speed comparison is invalid: the ordinary cache-producing forward was lazy and completed after its request timer started, whereas `encode_native_memory` calls `mx.eval` before native request timing. The corrected scaling benchmark synchronizes both representations and reports cold/cold separately from warm/warm. The selector-frozen online E0/E2 benchmark remains the serving-path economic reference.

### 13.2 Routed evidence rather than oracle evidence

We next remove the oracle ordering. Each HotpotQA and 2Wiki article, or each QASPER paragraph, becomes a typed `AgentResource`. The product SDK’s `PersistentResourceIndex` combines normalized token overlap, indexed BM25, and a deterministic signed-hash semantic control. It ranks the original question against all candidates; the top four are joined in rank order, truncated to the same 384-token budget, and encoded as native K/V. This is a parameter-free product-path baseline, not a learned retriever. Each dataset contains 20 distinct examples sampled under five seeds.

Table 18 exposes the discovery boundary. HotpotQA and 2Wiki have ten candidate documents per question and reach 0.725 and 0.775 recall@4. QASPER averages 46.6 paragraph candidates and reaches only 0.348. The selected sources consume 342–376 tokens. Mean scoring time is 2.85 ms on 2Wiki, 3.57 ms on HotpotQA, and

Table 18: Document routing before MLX materialization. Recall is the fraction of annotated evidence documents recovered in the first  $k$  ranks. Route time excludes one-time index construction.

| Dataset         | $n$ | Candidates | R@1   | R@2   | R@4   | Tokens | Route ms |
|-----------------|-----|------------|-------|-------|-------|--------|----------|
| qasper          | 20  | 46.6       | 0.119 | 0.140 | 0.348 | 370.6  | 11.92    |
| hotpotqa        | 20  | 10.0       | 0.400 | 0.600 | 0.725 | 376.3  | 3.57     |
| 2wikimultihopqa | 20  | 10.0       | 0.487 | 0.662 | 0.775 | 341.5  | 2.85     |

Table 19: Original-answer generation after routed materialization. Oracle uses annotation-scoped evidence; shuffled supplies another example’s routed K/V.

| Dataset         | Condition | $n$ | Token F1 | Gold log-prob. |
|-----------------|-----------|-----|----------|----------------|
| qasper          | Oracle    | 20  | 0.148    | -27.25         |
| qasper          | Routed    | 20  | 0.116    | -38.36         |
| qasper          | Shuffled  | 20  | 0.034    | -54.13         |
| qasper          | No memory | 20  | 0.026    | -60.20         |
| hotpotqa        | Oracle    | 20  | 0.129    | -10.95         |
| hotpotqa        | Routed    | 20  | 0.091    | -15.98         |
| hotpotqa        | Shuffled  | 20  | 0.054    | -20.30         |
| hotpotqa        | No memory | 20  | 0.058    | -15.68         |
| 2wikimultihopqa | Oracle    | 20  | 0.059    | -10.30         |
| 2wikimultihopqa | Routed    | 20  | 0.082    | -14.31         |
| 2wikimultihopqa | Shuffled  | 20  | 0.054    | -17.00         |
| 2wikimultihopqa | No memory | 20  | 0.073    | -13.88         |

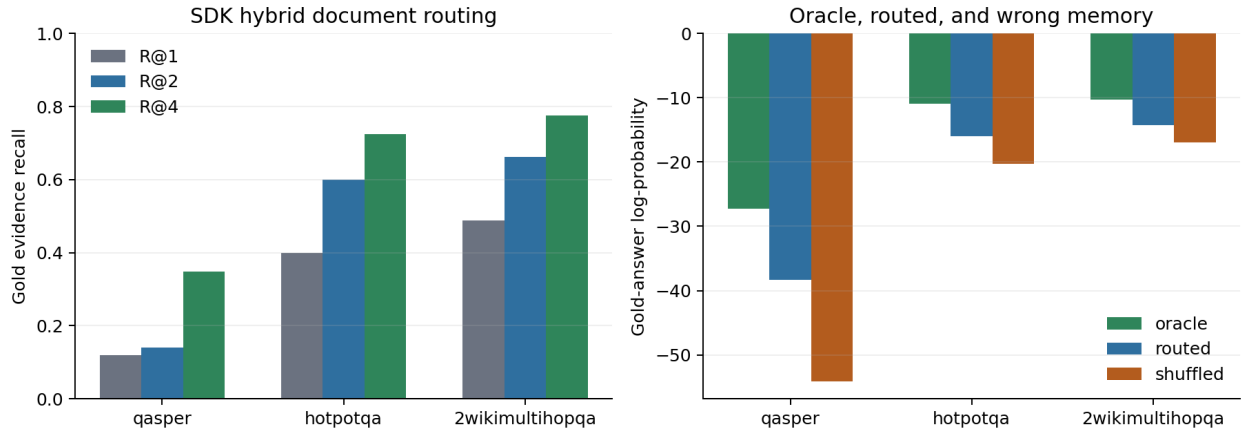


Figure 6: Left: evidence discovery improves as four candidates are admitted, but QASPER paragraph routing remains difficult. Right: routed memory carries more answer signal than shuffled memory, while the oracle gap remains.

11.92 ms on QASPER; corresponding one-time index construction means are 6.56, 8.20, and 32.98 ms. These small exhaustive Python cohorts establish cost accounting, not a production latency claim.

The routed ordinary-cache and native-K/V paths produce the same output and exactly equal gold-answer log-probability on all 60 pairs. MLX consumption therefore adds no measured quality error after selection. Routed memory is better than shuffled memory in both F1 and gold likelihood on every dataset. It remains below oracle likelihood, however: QASPER changes from  $-27.25$  to  $-38.36$ , HotpotQA from  $-10.95$  to  $-15.98$ , and 2Wiki from  $-10.30$  to  $-14.31$ . HotpotQA and 2Wiki routed likelihood also does not consistently beat the absent-memory control, despite better routed F1. Partial retrieval can be insufficient or misleading, while short free generation is noisy. The result closes the routed engine path and localizes the remaining deficit to evidence selection; it does not close a production QA-quality gate.

We then expand each dataset to 50 sampled examples (49 unique for QASPER and 50 for the other two). Table 20 confirms exact ordinary/native consumption parity and sharpens the dataset-specific result. On QASPER, routed native memory raises F1 from 0.038 without memory to 0.113, while shuffled memory reaches 0.042; on HotpotQA the corresponding values are 0.054, 0.113, and 0.034. Their gold-answer likelihoods move in the same causal direction. On 2Wiki, however, routed F1 is 0.076 versus 0.081 without memory and 0.089 shuffled, although routed likelihood remains better than shuffled. The larger cohort therefore confirms QASPER/Hotpot utility and a 2Wiki selection or answer-metric failure; it does not support a dataset-averaged quality claim.

Table 20: Expanded routed QA. Selection is frozen before ordinary or native consumption; oracle and shuffled conditions are causal bounds.

| Dataset  | Condition       | Sampled/unique | Recall@4 | F1    | $\log p(y)$ | Route ms |
|----------|-----------------|----------------|----------|-------|-------------|----------|
| QASPER   | no_memory       | 50/49          | 0.343    | 0.038 | -51.37      | 14.27    |
| QASPER   | oracle_native   | 50/49          | 0.343    | 0.162 | -22.83      | 14.27    |
| QASPER   | routed_native   | 50/49          | 0.343    | 0.113 | -36.13      | 14.27    |
| QASPER   | routed_ordinary | 50/49          | 0.343    | 0.113 | -36.13      | 14.27    |
| QASPER   | routed_shuffled | 50/49          | 0.343    | 0.042 | -46.76      | 14.27    |
| HotpotQA | no_memory       | 50/50          | 0.750    | 0.054 | -17.99      | 4.17     |
| HotpotQA | oracle_native   | 50/50          | 0.750    | 0.134 | -11.09      | 4.17     |
| HotpotQA | routed_native   | 50/50          | 0.750    | 0.113 | -14.30      | 4.17     |
| HotpotQA | routed_ordinary | 50/50          | 0.750    | 0.113 | -14.30      | 4.17     |
| HotpotQA | routed_shuffled | 50/50          | 0.750    | 0.034 | -20.82      | 4.17     |
| 2Wiki    | no_memory       | 50/50          | 0.725    | 0.081 | -12.93      | 3.42     |
| 2Wiki    | oracle_native   | 50/50          | 0.725    | 0.084 | -11.53      | 3.42     |
| 2Wiki    | routed_native   | 50/50          | 0.725    | 0.076 | -14.12      | 3.42     |
| 2Wiki    | routed_ordinary | 50/50          | 0.725    | 0.076 | -14.12      | 3.42     |
| 2Wiki    | routed_shuffled | 50/50          | 0.725    | 0.089 | -15.22      | 3.42     |

### 13.3 Optional learned routing across MLX model families

The remaining discovery deficit motivates a controlled adapter comparison, but it does not justify replacing the parameter-free router by default. We freeze the normalized attention-input states produced by exact 4-bit MLX revisions of Qwen3-4B, Qwen3-14B, Llama-3.1-8B, and Gemma-3-1B. For each model, an asymmetric rank-128 linear router is trained on 48 QASPER/HotpotQA examples for 512 steps under five seeds. Selection uses combined validation  $AUC_{0-30}$ ; the 32 test identities remain held out. Generic cosine routing and learned routing then see the same chunks, queries, and model states.

The learned router improves QASPER  $R_{20}$  for every model, from 0.413 to 0.640 at Qwen3-4B, 0.318 to 0.679 at Qwen3-14B, 0.318 to 0.468 at Llama-3.1-8B, and 0.226 to 0.454 at Gemma-3-1B. It does not transfer uniformly: HotpotQA  $R_{20}$  is lower for all four selected routers. This result narrows rather than closes the discovery gap. Public bundles therefore retain generic cosine as the **balanced** profile and expose the adapter as **qasper-learned**. The comparison qualifies routing on exact MLX model identities; it does not alter the selector-frozen E0/E2 transport results or establish end-task generation quality.



Table 21: Held-out routing comparison for exact post-trained HF and 4-bit MLX model identities. Cohort sizes are retained in the source artifact. The learned router is selected from five seeds using validation data;  $R_{20}$  is evidence recall at a 20% chunk budget.

| Model                       | QASPER $R_{20}$ |         | HotpotQA $R_{20}$ |         | Combined MRR |         |
|-----------------------------|-----------------|---------|-------------------|---------|--------------|---------|
|                             | Generic         | Learned | Generic           | Learned | Generic      | Learned |
| Qwen2.5-1.5B-Instruct       | 0.296           | 0.381   | 0.385             | 0.334   | 0.354        | 0.423   |
| Qwen2.5-Coder-1.5B-Instruct | 0.173           | 0.394   | 0.398             | 0.215   | 0.346        | 0.412   |
| Qwen3-4B                    | 0.413           | 0.640   | 0.386             | 0.342   | 0.333        | 0.549   |
| Qwen3-14B                   | 0.318           | 0.679   | 0.494             | 0.314   | 0.401        | 0.645   |
| Llama-3.1-8B                | 0.318           | 0.468   | 0.616             | 0.421   | 0.404        | 0.537   |
| Gemma-3-1B                  | 0.226           | 0.454   | 0.336             | 0.318   | 0.316        | 0.494   |

### 13.4 Matched selected-text economics

The routed selections also support a direct E0/E2 comparison without changing the evidence. For 20 examples per dataset across five seeds, E0 restores an ordinary cache snapshot of the selected source and E2 restores its native PRA cache. The question, greedy decoder, and 24-token output budget are identical. The shared schedule includes cold one-shot, two warm repeats, three deterministic query formulations over the same resource, and eight threaded requests sharing one immutable cache object.

Table 22: Matched-selection quality on the shared three-engine cohort. All parity pools 840 request pairs per engine across four schedules.

| Engine     | Dataset         | Visible E0/E2 | Reduction | Cold F1 E0/E2 | Cold parity | All parity |
|------------|-----------------|---------------|-----------|---------------|-------------|------------|
| mlx-lm     | 2wikimultihopqa | 375/33        | 91.1%     | 0.067/0.067   | 100.0%      | 100.0%     |
| mlx-lm     | hotpotqa        | 415/39        | 90.6%     | 0.087/0.087   | 100.0%      | 100.0%     |
| mlx-lm     | qasper          | 399/28        | 93.0%     | 0.110/0.110   | 100.0%      | 100.0%     |
| sglang-mlx | 2wikimultihopqa | 375/33        | 91.1%     | 0.074/0.074   | 100.0%      | 100.0%     |
| sglang-mlx | hotpotqa        | 415/39        | 90.6%     | 0.084/0.084   | 100.0%      | 100.0%     |
| sglang-mlx | qasper          | 399/28        | 93.0%     | 0.105/0.105   | 100.0%      | 100.0%     |
| vllm-metal | 2wikimultihopqa | 378/33        | 91.2%     | 0.079/0.079   | 100.0%      | 100.0%     |
| vllm-metal | hotpotqa        | 417/39        | 90.6%     | 0.074/0.074   | 100.0%      | 100.0%     |
| vllm-metal | qasper          | 400/28        | 93.0%     | 0.101/0.101   | 100.0%      | 100.0%     |

Table 23: Macro E2/E0 execution-cost ratios; lower is better. Cold includes ingestion, concurrent is inverse requests/s, and E2 usable is milliseconds.

| Engine     | Cold  | Warm  | Multi-query | Concurrent | E2 usable ms |
|------------|-------|-------|-------------|------------|--------------|
| mlx-lm     | 1.130 | 1.155 | 1.148       | 1.132      | 51.2         |
| sglang-mlx | 0.976 | 1.136 | 1.140       | 1.122      | 55.0         |
| vllm-metal | 0.985 | 1.004 | 0.993       | 0.981      | 86.0         |

MLX-LM produces exactly the same E0 and E2 output in all 840 pairs, with equal F1 and gold evidence by construction. E2 reduces visible prompt tokens by 90.6–93.0%. Its mean time to usable context is 51.2 ms, but cold end-to-end cost is 1.130 times E0 because the unfused selected cache slows generation. Warm and multi-query ratios are 1.155 and 1.148. Under eight threaded requests, E2 reaches 88.4% of E0 throughput, an inverse cost ratio of 1.132. This is exact semantic transport across all four regimes, but not yet an optimized native-attention kernel.

The expanded 149-question confirmation preserves all 2,086 MLX pairs, with an engine-level Wilson 95% lower bound of 0.9982. Visible-token reduction remains 90.4–93.0%. Macro cold, warm, multi-query, and concurrent cost ratios are 1.043, 1.139, 1.133, and 1.072. The larger cohort thus reduces the measured cold and concurrent penalty relative to the original 20-example-per-dataset run, but confirms that this unfused implementation remains slower than E0 in every regime.

We isolate the principal decode allocation with a one-normalization segmented attention primitive. Rather than

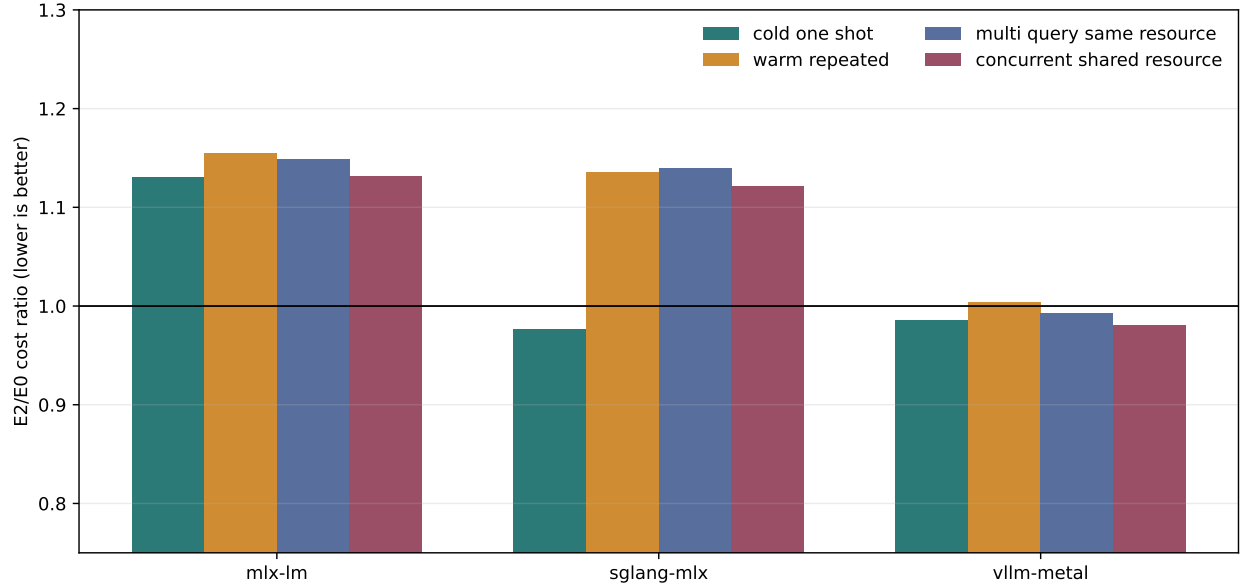


Figure 7: Macro representation cost after freezing retrieval. Lower is better; concurrent cost is inverse throughput. The current unfused MLX native cache is semantically exact but slower in every schedule.

concatenate selected and local K/V, it forms their score blocks separately, shares one maximum and denominator, and sums the two value numerators. We additionally graph-compile that primitive while keeping dynamic causal masks outside the captured graph. Table 24 reports paired M4 Pro kernel measurements with 384 selected tokens, eight heads, and 128-dimensional heads.

Table 24: Eager and graph-compiled segmented attention against concatenation. Ratios below one are faster. Avoided MiB is the temporary concatenated K/V per layer-call. Compilation is an MLX graph optimization, not a custom Metal kernel.

| Local tokens | concat ms | eager/concat | compiled/concat | avoided MiB | max error |
|--------------|-----------|--------------|-----------------|-------------|-----------|
| 2,048        | 0.351     | 1.053        | 1.099           | 9.5         | 1.22e-04  |
| 8,192        | 0.712     | 1.040        | 1.016           | 33.5        | 3.42e-05  |
| 32,768       | 2.135     | 0.990        | 0.959           | 129.5       | 3.05e-05  |

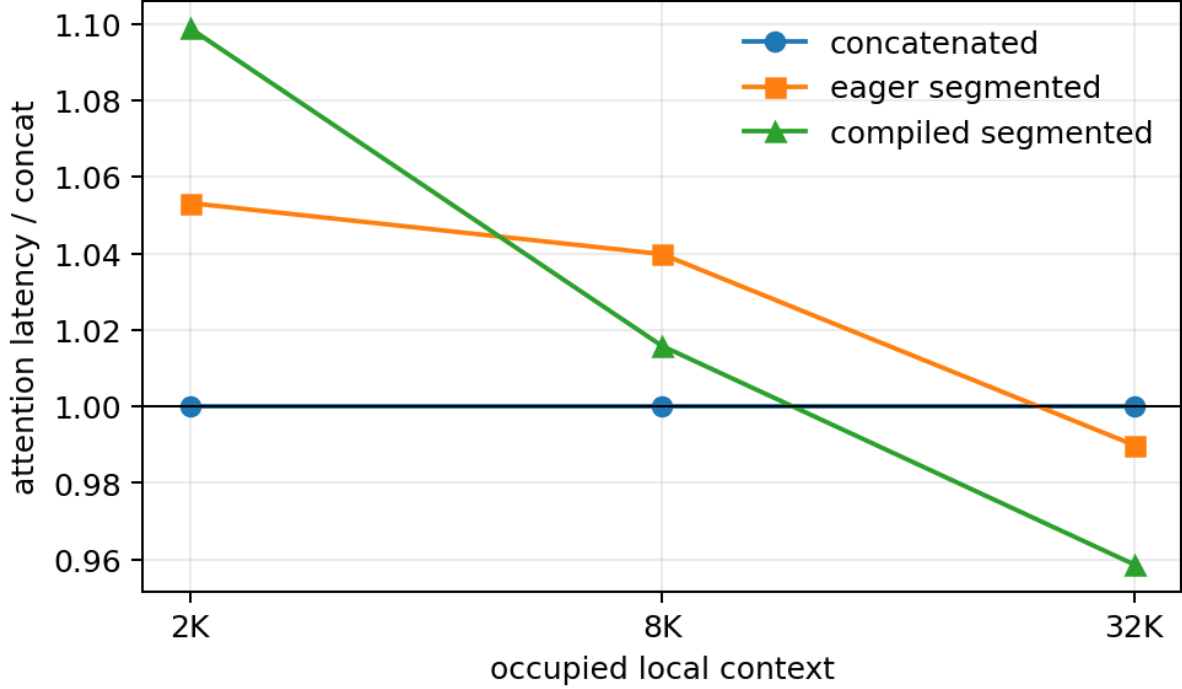


Figure 8: Segmented-attention kernel cost and temporary-memory avoidance as the occupied local context grows. Graph compilation reaches 0.959 times concat at 32K while avoiding 129.5 MiB per layer-call, but does not remove the full-model request overhead.

The compiled kernel is slower at 2K (1.099 times concat), near parity at 8K (1.016), and faster at 32K (0.959), with maximum absolute error at most  $1.22 \times 10^{-4}$ . The end-to-end check is less favorable. On the same 15 Qwen3-32B natural-QA examples, compiled concatenated E2 is 1.009 times E0 and preserves all outputs; compiled segmented E2 is 1.052 times E0 and agrees on 13/15 outputs. The eager ratios on that subset are 1.012 and 1.048. Graph capture therefore removes neither the residual model-wide overhead nor the reduction-order drift. A genuinely fused Metal kernel remains an open systems optimization.

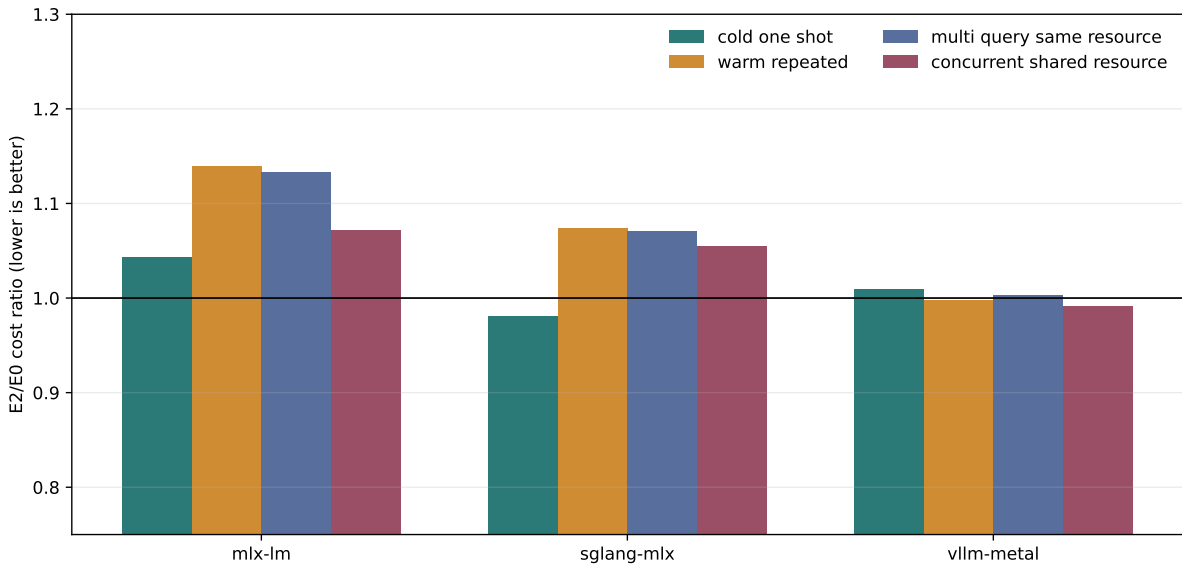


Figure 9: Expanded 149-question representation-cost confirmation across all three engines. Retrieval and selected evidence are held fixed.

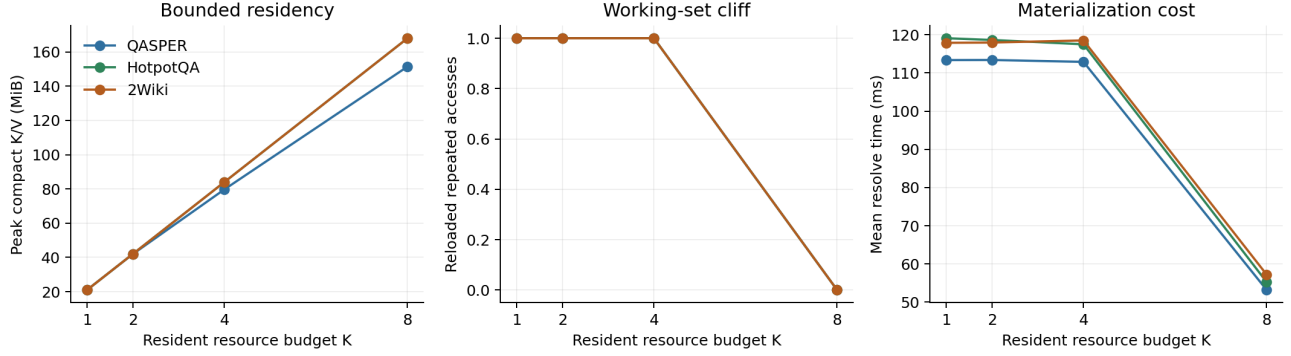


Figure 10: Five-seed compact-K/V pressure over QASPER, HotpotQA, and 2Wiki. Each seed visits eight resources twice and then revisits the first. The budget  $K = 8$  is the first condition that fits the working set.

Table 25: Natural-QA residency curve. Each row contains 85 requests: 17 requests per seed over five seeds. Reload fraction divides reloads by the nine accesses after each resource’s first load.

| Dataset  | $K$ | $n$ | Peak MiB | Reload frac. | Resolve ms | Token F1 |
|----------|-----|-----|----------|--------------|------------|----------|
| QASPER   | 1   | 85  | 21.0     | 1.00         | 113.4      | 0.189    |
| QASPER   | 2   | 85  | 41.9     | 1.00         | 113.4      | 0.189    |
| QASPER   | 4   | 85  | 79.7     | 1.00         | 112.9      | 0.189    |
| QASPER   | 8   | 85  | 151.4    | 0.00         | 53.2       | 0.189    |
| HotpotQA | 1   | 85  | 21.0     | 1.00         | 119.1      | 0.144    |
| HotpotQA | 2   | 85  | 42.0     | 1.00         | 118.6      | 0.144    |
| HotpotQA | 4   | 85  | 84.0     | 1.00         | 117.5      | 0.144    |
| HotpotQA | 8   | 85  | 168.0    | 0.00         | 55.2       | 0.144    |
| 2Wiki    | 1   | 85  | 21.0     | 1.00         | 117.9      | 0.049    |
| 2Wiki    | 2   | 85  | 42.0     | 1.00         | 117.9      | 0.049    |
| 2Wiki    | 4   | 85  | 84.0     | 1.00         | 118.5      | 0.049    |
| 2Wiki    | 8   | 85  | 168.0    | 0.00         | 57.2       | 0.049    |

Figure 10 and Table 25 validate physical lifecycle rather than theoretical byte sums. Each request pins one selected block, dequantizes it locally, generates, and releases it. With budgets  $K \in \{1, 2, 4\}$ , the cyclic eight-resource working set reloads all nine repeated accesses per seed. At  $K = 8$ , initial loads fall from 17 to eight, evictions and reloads fall to zero, and mean resolution falls from 113–119 ms to 53–57 ms. Peak compact residency rises from approximately 21 MiB at  $K = 1$  to 151 MiB on QASPER and 168 MiB on HotpotQA and 2Wiki at  $K = 8$ .

Token F1 is unchanged across all four budgets within every dataset, and mean completion latency remains within roughly 5 ms because generation dominates the saved resolution time. The result is therefore a working-set threshold, not evidence that more residency generally improves answer quality or complete request latency. No pinned block is evicted and compact detail never enters an ordinary prompt or rotating cache.

### 13.5 Larger-model sustained pressure on M4 Pro

We repeat the bounded-residency experiment with Qwen3-4B on the M4 Pro. Each seed owns eight resources, visits the complete set three times, and revisits the first resource. Four resident budgets produce 25 requests per seed, or 125 requests per table row and 1,500 requests overall.

Table 26: Qwen3-4B sustained compact-K/V pressure on the M4 Pro. Budget is resident/logical resources. Resolve includes compact-detail load and dequantization; p95 retains unavoidable first-ingestion requests.

| Dataset  | Budget | $n$ | F1    | Reload | Resolve ms | p95 ms | Resident MiB |
|----------|--------|-----|-------|--------|------------|--------|--------------|
| QASPER   | 1/8    | 125 | 0.218 | 0.680  | 417.1      | 460.6  | 24.4         |
| QASPER   | 2/8    | 125 | 0.218 | 0.680  | 420.0      | 460.7  | 47.7         |
| QASPER   | 4/8    | 125 | 0.218 | 0.680  | 419.3      | 460.5  | 91.5         |
| QASPER   | 8/8    | 125 | 0.218 | 0.000  | 134.3      | 460.2  | 167.4        |
| HotpotQA | 1/8    | 125 | 0.231 | 0.680  | 457.2      | 460.2  | 27.0         |
| HotpotQA | 2/8    | 125 | 0.231 | 0.680  | 457.8      | 461.0  | 52.9         |
| HotpotQA | 4/8    | 125 | 0.231 | 0.680  | 457.6      | 461.1  | 101.5        |
| HotpotQA | 8/8    | 125 | 0.231 | 0.000  | 146.5      | 460.1  | 185.8        |
| 2Wiki    | 1/8    | 125 | 0.128 | 0.680  | 458.8      | 467.1  | 27.0         |
| 2Wiki    | 2/8    | 125 | 0.128 | 0.680  | 461.1      | 478.1  | 52.9         |
| 2Wiki    | 4/8    | 125 | 0.128 | 0.680  | 462.2      | 480.6  | 101.5        |
| 2Wiki    | 8/8    | 125 | 0.128 | 0.000  | 147.1      | 465.8  | 185.8        |

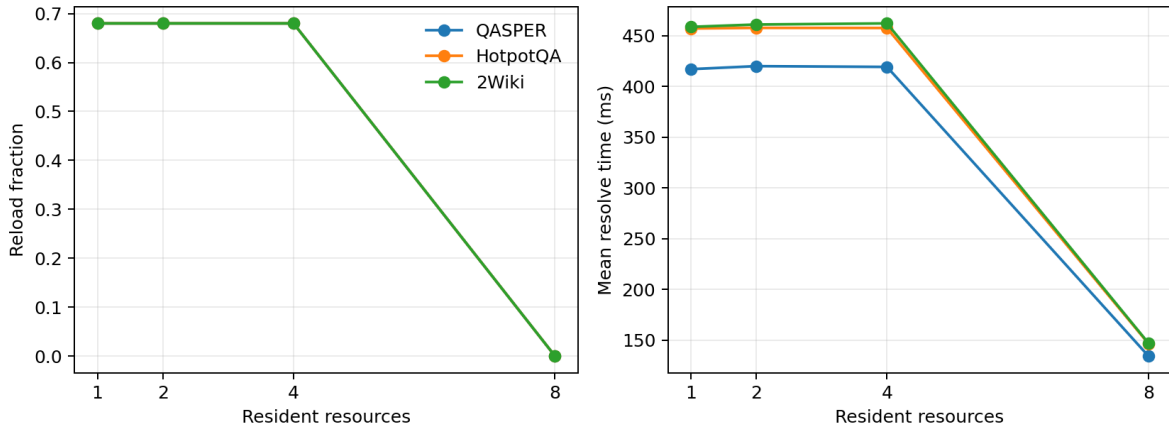


Figure 11: The eight-resource working-set threshold at Qwen3-4B. Fitting the complete set removes repeat reloads and lowers mean resolution; it does not remove each resource’s first cold ingestion.

For budgets 1, 2, and 4, 68.0% of all requests reload compact detail. At budget 8, reloads fall to zero. Mean resolution falls from 417.1 to 134.3 ms on QASPER, 457.2 to 146.5 ms on HotpotQA, and 458.8 to 147.1 ms on 2Wiki, a 67.8–68.0% reduction. Mean compact residency rises to 167.4 MiB on QASPER and 185.8 MiB on HotpotQA and 2Wiki. Token F1 and gold-answer log-probability are identical across all four budgets within each dataset.

The p95 resolve time remains 460–466 ms at budget 8 because eight of each 25-request sequence are compulsory first loads. Residency removes 17 repeat loads, not those cold ingestions. This result therefore supports working-set admission and reuse policy; it does not support a claim that larger budgets improve answer quality or cold-start tails.

### 13.6 Shared storage lifecycle and MLX realization

The product runtime now defines engine-neutral HOT/WARM/COLD/SOURCE semantics. MLX attention-ready arrays are HOT. Lossless retained arrays or strict fingerprinted files are WARM. Compressed persistence is COLD, with K/V quantization configured independently from compression. Apple unified memory can physically host both HOT and WARM, but readiness and lifecycle remain different service states.

Across the common four-label, five-seed policy workload, lossless tier recovery and SOURCE reconstruction are exact in 20/20 runs. Weighted retention keeps the shared document where plain LRU does not; active-task and dependency guards hold in 20/20, and a grace period avoids 20/20 transient writes. A generic per-array symmetric-int8 codec occupies 0.25 of FP32 bytes with 0.0103 RMSE. That codec result is not an MLX answer-quality claim. The model-native per-head experiment in this paper is the applicable quality evidence and obtains a roughly twofold compact-residency reduction because its metadata and scale layout differ. COLD still dequantizes before attention, so compact persistence does not reduce active attention bytes.

The live storage probe now replaces opaque WARM blobs with independently checksummed, mmap-readable K and V segments per layer. Three QASPER selections generate from MLX HOT arrays, lossless WARM reload, int8 COLD reload, and a fresh manager after restart. WARM reproduces all 3/3 HOT outputs and restart recovery succeeds. Median promotion-plus-generation latency is 327 ms HOT and 644 ms WARM (1.97 $\times$ ); the excluded background WARM transition costs 348 ms. COLD promotion plus generation is 600 ms and its background conversion costs 940 ms.

Table 27: Shared three-example live storage probe. Prefix is mean HOT/int8 common output prefix; ratios use lifecycle-inclusive request latency.

| Engine     | WARM exact | int8 exact | int8 first | Prefix | WARM/HOT | COLD/HOT |
|------------|------------|------------|------------|--------|----------|----------|
| mlx-lm     | 100%       | 0%         | 67%        | 1.3    | 1.97     | 1.84     |
| sglang-mlx | 100%       | 0%         | 67%        | 0.7    | 1.90     | 1.89     |
| vllm-metal | 100%       | 0%         | 67%        | 4.0    | 1.96     | 2.00     |

Per-array int8 COLD changes all three 24-token sequences and preserves the first token in two. Its positive mean F1 change (+0.068) is an unstable three-example accident rather than evidence of improved quality: exact parity is zero and the mean common prefix is 1.3 tokens. Lossless persistence is the default; quantized COLD requires a larger model- and workload-specific quality gate. The separate per-head compact-residency experiment remains relevant but does not override this autoregressive sequence diagnostic.

We expand the protocol to 80 examples: all three datasets at Qwen3-0.6B and QASPER at Qwen3-1.7B. Lossless WARM remains 80/80 exact and all four managers recover after restart; int8 COLD is exact in 13/80 sequences. WARM/HOT p50 ratios are 2.01–2.15, with WARM p95 between 710 and 817 ms. The larger cohort therefore strengthens persistence correctness while making request-time promotion, rather than array storage, the immediate systems bottleneck. Two five-example QASPER replications add Llama-3.2-1B and Gemma-3-1B; both are 5/5 lossless-WARM exact with restart recovery. Llama int8 is 2/5 exact and Gemma is 5/5 in this small cohort, which is insufficient to override the broader Qwen-based opt-in gate.

### 13.7 Asynchronous promotion and online request behavior

The lifecycle ratios above charge a complete WARM promotion to the requesting thread. We therefore move promotion ownership to the event loop, deduplicate concurrent promotions by logical key, admit completed objects to a bounded hot set, and overlap promotion with query-side work. Table 29 and Figure 12 report five QASPER

Table 28: Expanded MLX lifecycle cohorts. Latencies include synchronous promotion and generation; int8 remains diagnostic.

| Engine | Model                      | Data            | $n$ | WARM exact | int8 exact | HOT p50 | WARM p50 | WARM p95 |
|--------|----------------------------|-----------------|-----|------------|------------|---------|----------|----------|
| mlx-lm | Llama-3.2-1B-Instruct-4bit | qasper          | 5   | 100.0%     | 40.0%      | 191     | 275      | 281      |
| mlx-lm | Qwen3-0.6B-4bit            | 2wikimultihopqa | 20  | 100.0%     | 15.0%      | 328     | 658      | 710      |
| mlx-lm | Qwen3-0.6B-4bit            | hotpotqa        | 20  | 100.0%     | 15.0%      | 329     | 707      | 817      |
| mlx-lm | Qwen3-0.6B-4bit            | qasper          | 20  | 100.0%     | 15.0%      | 326     | 672      | 714      |
| mlx-lm | Qwen3-1.7B-4bit            | qasper          | 20  | 100.0%     | 20.0%      | 337     | 723      | 744      |
| mlx-lm | gemma-3-1b-it-4bit         | qasper          | 5   | 100.0%     | 100.0%     | 391     | 480      | 502      |

selections at each lead. Every generation remains HOT-exact. With no lead, median demand stall is 383.7 ms and demand-path cost is 2.123 times HOT. A 250 ms lead lowers these to 90.3 ms and 1.310; 350 ms makes 40% ready at demand and reaches 1.048; 500 ms makes all five ready and reaches 0.983. Promotion is therefore hideable when the scheduler has enough useful work, but it has not become intrinsically cheaper.

Table 29: Event-loop-owned WARM promotion. Lead and stall are milliseconds.

| Engine | Lead   | Ready | Stall p50 | Stall p95 | Demand/HOT |
|--------|--------|-------|-----------|-----------|------------|
| mlx    | 0 ms   | 0%    | 383.7     | 397.6     | 2.123      |
| mlx    | 25 ms  | 0%    | 332.7     | 358.1     | 2.054      |
| mlx    | 50 ms  | 0%    | 304.3     | 328.3     | 1.962      |
| mlx    | 100 ms | 0%    | 256.4     | 305.5     | 1.836      |
| mlx    | 250 ms | 0%    | 90.3      | 122.4     | 1.310      |
| mlx    | 350 ms | 40%   | 4.2       | 40.8      | 1.048      |
| mlx    | 500 ms | 100%  | 0.1       | 0.1       | 0.983      |
| sglang | 0 ms   | 0%    | 210.3     | 280.1     | 1.561      |
| sglang | 25 ms  | 0%    | 202.4     | 240.7     | 1.495      |
| sglang | 50 ms  | 0%    | 169.0     | 195.7     | 1.475      |
| sglang | 100 ms | 0%    | 111.9     | 141.4     | 1.326      |
| sglang | 250 ms | 80%   | 0.0       | 5.7       | 1.035      |
| sglang | 350 ms | 100%  | 0.0       | 0.1       | 1.063      |
| sglang | 500 ms | 100%  | 0.0       | 0.0       | 1.023      |

The native executor is also exercised through the OpenAI-compatible gateway, including streaming, cancellation, cleanup, and session ownership. A shared gateway correction normalizes native token rows into portable deltas and a terminal completion event. At concurrency one through sixteen, throughput is 1.52–1.70 requests/s; TTFT p95 rises from 302 to 9,988 ms as the deliberately serialized model runner queues. Every one of the 31 concurrent outputs matches the prime output. The client cancels after the first streamed token (364 ms) and all 34 exercised sessions close without retaining request-owned native detail. This is a real HTTP-path measurement, but the single MLX model instance is intentionally serialized to avoid unsafe concurrent mutation of prompt-cache arrays.

### 13.8 Concurrent physical-tier curves

The lifecycle manager is next stressed at concurrency  $c \in \{1, 2, 4, 8, 16\}$  with either one shared immutable resource or one resource per request. The model runner remains deliberately serialized, so these finite waves measure queueing, promotion, and physical sharing rather than continuous batching. All 124 HOT/WARM requests reproduce their HOT baselines exactly. At  $c = 16$ , shared HOT and WARM sustain 3.20 and 2.91 requests/s with request p95 of 4,996 and 5,491 ms. The shared WARM wave promotes one object and avoids 630 MiB of duplicate physical K/V. Independent WARM instead promotes 16 objects, reads 1.26 GiB, reaches 1.22 requests/s, and has 13,087 ms p95. The gap isolates resource sharing and promotion amplification; it is not evidence for parallel model execution. Int8 COLD remains outside the lossless gate.

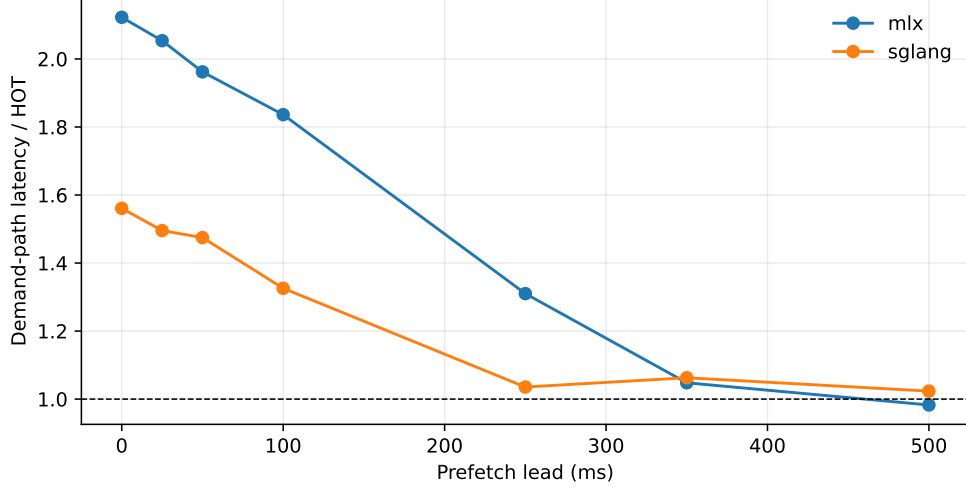


Figure 12: Promotion lead trades demand-path stall for speculative work. The dashed line is HOT request latency.

Table 30: Online native-gateway concurrency, including queueing. Times are ms.

| Engine     | $c$ | req/s | TTFT p50 | TTFT p95 | request p50 | request p95 |
|------------|-----|-------|----------|----------|-------------|-------------|
| mlx-lm     | 1   | 1.70  | 302      | 302      | 589         | 589         |
| mlx-lm     | 2   | 1.52  | 355      | 1028     | 642         | 1316        |
| mlx-lm     | 4   | 1.66  | 950      | 2138     | 1233        | 2415        |
| mlx-lm     | 8   | 1.58  | 2247     | 4779     | 2538        | 5066        |
| mlx-lm     | 16  | 1.56  | 4838     | 9988     | 5123        | 10275       |
| sglang-mlx | 1   | 3.06  | 36       | 36       | 325         | 325         |
| sglang-mlx | 2   | 2.94  | 43       | 381      | 353         | 678         |
| sglang-mlx | 4   | 3.06  | 369      | 1007     | 670         | 1305        |
| sglang-mlx | 8   | 3.00  | 1037     | 2364     | 1345        | 2667        |
| sglang-mlx | 16  | 2.99  | 2337     | 5027     | 2619        | 5326        |

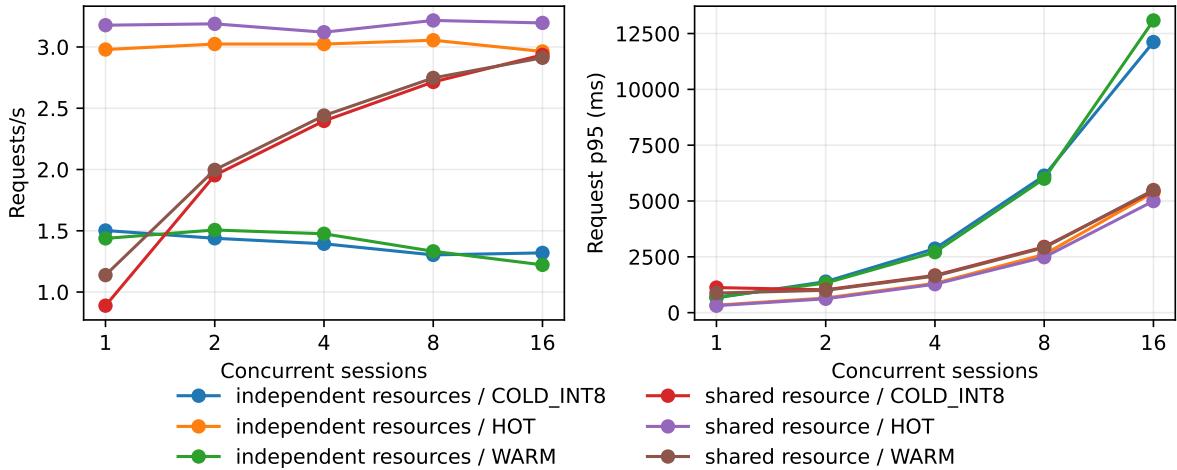


Figure 13: MLX physical-tier concurrency. Shared WARM amortizes one promotion; independent WARM scales storage work with the number of resources.



### 13.9 Selective quantization and sustained pressure

Twenty QASPER examples isolate which tensors can tolerate int8 COLD storage. Quantizing every key and value is sequence-exact in 4/20 cases. Keys alone are 5/20; values alone are 10/20; the late half is 11/20; and the late quarter is 14/20, with first-token parity in all 20 and a mean 18.85-token common prefix. Aggregate F1 changes remain small, but autoregressive divergence makes even the late-quarter profile a calibration candidate rather than a default.

A separate sustained-session study issues 1,125 requests over QASPER, HotpotQA, and 2Wiki: five seeds, eight resources, three complete rounds and a final revisit, under HOT capacities two, four, and eight. Capacities two and four average 17 reloads per seed and reload the final revisit in every case; capacity eight has no reload or eviction. F1 and answer log-probability are identical across budgets within each dataset. The intervention changes physical reuse, not selected evidence or model semantics.

Table 33: Long-session pressure. Reloads and evictions are means over five seeds.

| Dataset         | HOT resources | Requests | Reloads | Evictions | Final reload | F1    | p95 ms |
|-----------------|---------------|----------|---------|-----------|--------------|-------|--------|
| qasper          | 2             | 125      | 17.0    | 23.0      | 100%         | 0.172 | 493    |
| qasper          | 4             | 125      | 17.0    | 21.0      | 100%         | 0.172 | 493    |
| qasper          | 8             | 125      | 0.0     | 0.0       | 0%           | 0.172 | 495    |
| hotpotqa        | 2             | 125      | 17.0    | 23.0      | 100%         | 0.131 | 497    |
| hotpotqa        | 4             | 125      | 17.0    | 21.0      | 100%         | 0.131 | 500    |
| hotpotqa        | 8             | 125      | 0.0     | 0.0       | 0%           | 0.131 | 501    |
| 2wikimultihopqa | 2             | 125      | 17.0    | 23.0      | 100%         | 0.064 | 497    |
| 2wikimultihopqa | 4             | 125      | 17.0    | 21.0      | 100%         | 0.064 | 508    |
| 2wikimultihopqa | 8             | 125      | 0.0     | 0.0       | 0%           | 0.064 | 502    |

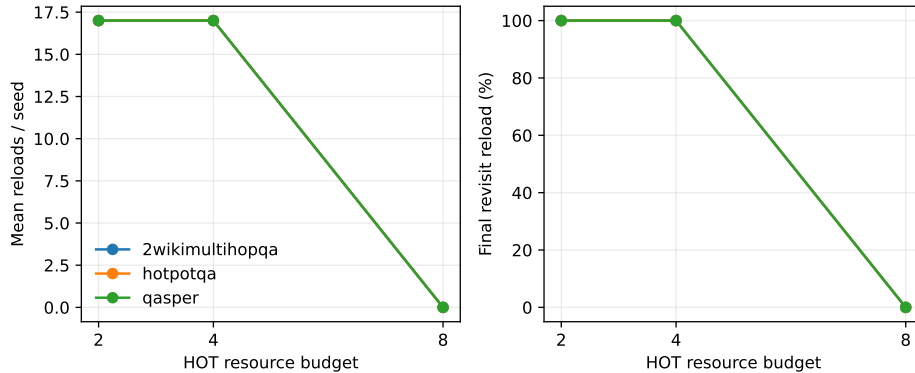


Figure 14: The eight-resource working-set threshold appears consistently across all three datasets.

The final pressure experiment crosses physical HOT and WARM budgets with local rotating windows of 64 and 256 tokens over 1,000 QASPER requests. With HOT/WARM = 2/2, 75.2% of requests reconstruct from SOURCE; raising WARM to eight removes SOURCE starts. With HOT/WARM = 8/2, starts are 68% HOT, 8% WARM, and 24% SOURCE; at 8/8, the latter two become 32% WARM and zero SOURCE. F1 is 0.123 in every cell, and changing the local window does not materially move completion p95. Disk-backed WARM also fails to improve resolve p95 over this mixture of small-source reconstruction: retaining all eight objects reaches 400–429 ms, while the tighter WARM budget reaches 347–398 ms. The product implication is cost-aware admission, not maximal persistence.

The deterministic admission calculation makes that implication concrete. Across this pressure trace, SOURCE reconstruction averages 335.7 ms and WARM restore averages 346.0 ms; creating the WARM representation costs another 396.8 ms on average. Thus

$$r(C_{\text{source}} - C_{\text{warm}}) > C_{\text{persist}}$$

has no positive break-even reuse count on this cohort. The correct default for these small reconstructable QASPER records is SOURCE, not WARM. This is a hardware- and representation-specific decision, not a rejection of

Table 31: Complete finite-wave storage-concurrency results. Times are ms; exact/HOT is the transport-fidelity gate.

| Workload              | Tier      | c  | req/s | p50  | p95   | p99   | queue p95 | exact/HOT |
|-----------------------|-----------|----|-------|------|-------|-------|-----------|-----------|
| shared_resource       | hot       | 1  | 3.18  | 313  | 313   | 313   | 0         | 100.0%    |
| shared_resource       | hot       | 2  | 3.19  | 314  | 621   | 621   | 308       | 100.0%    |
| shared_resource       | hot       | 4  | 3.12  | 626  | 1274  | 1274  | 944       | 100.0%    |
| shared_resource       | hot       | 8  | 3.22  | 1226 | 2484  | 2484  | 2165      | 100.0%    |
| shared_resource       | hot       | 16 | 3.20  | 2470 | 4996  | 4996  | 4688      | 100.0%    |
| independent_resources | hot       | 1  | 2.98  | 335  | 335   | 335   | 0         | 100.0%    |
| independent_resources | hot       | 2  | 3.02  | 331  | 657   | 657   | 328       | 100.0%    |
| independent_resources | hot       | 4  | 3.02  | 666  | 1313  | 1313  | 997       | 100.0%    |
| independent_resources | hot       | 8  | 3.06  | 1328 | 2607  | 2607  | 2274      | 100.0%    |
| independent_resources | hot       | 16 | 2.96  | 2781 | 5384  | 5384  | 5082      | 100.0%    |
| shared_resource       | warm      | 1  | 1.14  | 877  | 877   | 877   | 0         | 100.0%    |
| shared_resource       | warm      | 2  | 2.00  | 710  | 1001  | 1001  | 710       | 100.0%    |
| shared_resource       | warm      | 4  | 2.44  | 999  | 1638  | 1638  | 1318      | 100.0%    |
| shared_resource       | warm      | 8  | 2.75  | 1648 | 2909  | 2909  | 2593      | 100.0%    |
| shared_resource       | warm      | 16 | 2.91  | 2950 | 5491  | 5491  | 5176      | 100.0%    |
| independent_resources | warm      | 1  | 1.44  | 695  | 695   | 695   | 0         | 100.0%    |
| independent_resources | warm      | 2  | 1.51  | 723  | 1327  | 1327  | 723       | 100.0%    |
| independent_resources | warm      | 4  | 1.48  | 1377 | 2710  | 2710  | 1963      | 100.0%    |
| independent_resources | warm      | 8  | 1.33  | 2938 | 5999  | 5999  | 5246      | 100.0%    |
| independent_resources | warm      | 16 | 1.22  | 6673 | 13086 | 13086 | 12310     | 100.0%    |
| shared_resource       | cold_int8 | 1  | 0.89  | 1123 | 1123  | 1123  | 0         | 0.0%      |
| shared_resource       | cold_int8 | 2  | 1.95  | 708  | 1024  | 1024  | 708       | 0.0%      |
| shared_resource       | cold_int8 | 4  | 2.40  | 1037 | 1668  | 1668  | 1358      | 0.0%      |
| shared_resource       | cold_int8 | 8  | 2.71  | 1686 | 2945  | 2945  | 2640      | 0.0%      |
| shared_resource       | cold_int8 | 16 | 2.94  | 2901 | 5446  | 5446  | 5125      | 0.0%      |
| independent_resources | cold_int8 | 1  | 1.50  | 665  | 665   | 665   | 0         | 0.0%      |
| independent_resources | cold_int8 | 2  | 1.44  | 715  | 1389  | 1389  | 715       | 0.0%      |
| independent_resources | cold_int8 | 4  | 1.39  | 1403 | 2867  | 2867  | 2092      | 0.0%      |
| independent_resources | cold_int8 | 8  | 1.30  | 3173 | 6135  | 6135  | 5373      | 0.0%      |
| independent_resources | cold_int8 | 16 | 1.32  | 6179 | 12119 | 12119 | 11329     | 12.5%     |

Table 32: Selective int8 COLD calibration on 20 QASPER examples.

| Profile         | Exact  | First  | Prefix | $\Delta F1$ | $\Delta \log p$ |
|-----------------|--------|--------|--------|-------------|-----------------|
| lossless        | 100.0% | 100.0% | 24.0   | +0.0000     | +0.0000         |
| all_kv          | 20.0%  | 95.0%  | 8.8    | -0.0163     | +0.2862         |
| all_keys        | 25.0%  | 85.0%  | 9.8    | -0.0039     | +0.0905         |
| all_values      | 50.0%  | 100.0% | 14.6   | +0.0121     | +0.1798         |
| early_half_kv   | 15.0%  | 85.0%  | 8.6    | +0.0012     | +0.2602         |
| late_half_kv    | 55.0%  | 95.0%  | 16.6   | -0.0028     | +0.0771         |
| late_quarter_kv | 70.0%  | 100.0% | 18.9   | -0.0031     | +0.0698         |

persistent K/V for expensive encoders or remote sources.

The requested 2K, 8K, 32K, and full local-history frontier is not inferred from the 64/256-token cache-capacity control. Those cells require prompts that actually occupy each window. We therefore label the current joint frontier partial: resource budgets 2/8, concurrency 1–16, and consumer-layer profiles are measured, while long local-window occupancy remains a separate staged run. The last-half consumer profile preserves 5/5 controlled recoveries at half the active selected K/V bytes; last-quarter and last-four do not. The larger natural ladder also rejects last-half consumption, so none is a BALANCED or ECONOMY recommendation.

Table 34: Crossed physical-tier and local-window pressure. Start percentages name the representation available before each request. Times are ms.

| HOT | WARM | Window | Requests | HOT start | WARM start | SOURCE start | F1    | Resolve p95 | Total p95 |
|-----|------|--------|----------|-----------|------------|--------------|-------|-------------|-----------|
| 2   | 2    | 64     | 125      | 0%        | 25%        | 75%          | 0.123 | 398.2       | 354       |
| 2   | 2    | 256    | 125      | 0%        | 25%        | 75%          | 0.123 | 377.9       | 353       |
| 2   | 8    | 64     | 125      | 0%        | 100%       | 0%           | 0.123 | 428.6       | 353       |
| 2   | 8    | 256    | 125      | 0%        | 100%       | 0%           | 0.123 | 428.4       | 354       |
| 8   | 2    | 64     | 125      | 68%       | 8%         | 24%          | 0.123 | 351.9       | 345       |
| 8   | 2    | 256    | 125      | 68%       | 8%         | 24%          | 0.123 | 347.2       | 348       |
| 8   | 8    | 64     | 125      | 68%       | 32%        | 0%           | 0.123 | 402.8       | 346       |
| 8   | 8    | 256    | 125      | 68%       | 32%        | 0%           | 0.123 | 400.0       | 349       |

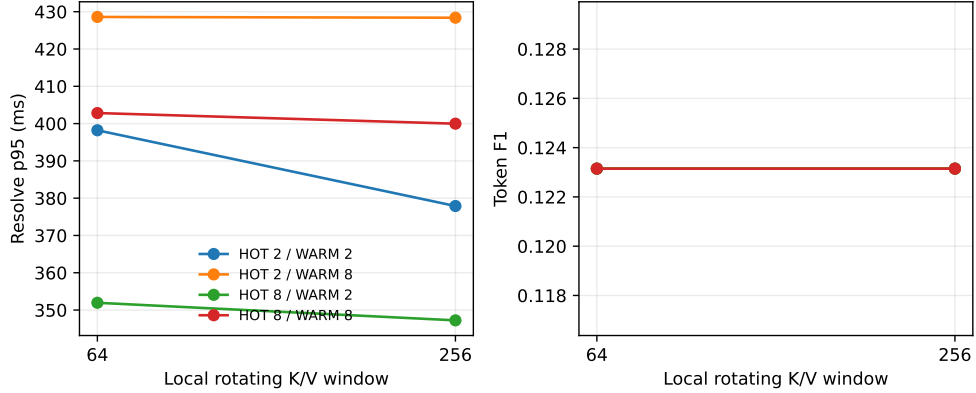


Figure 15: Local-window size has no quality effect in this bounded decode; HOT admission dominates median resolution, while broad WARM retention does not improve the tail on this host.

## 14 Promotion Gates

| Gate                      | Status               | Evidence                                                                                                  |
|---------------------------|----------------------|-----------------------------------------------------------------------------------------------------------|
| environment               | closed               | MLX Metal generation and server repeat reliably                                                           |
| prompt-cache coexistence  | smoke closed         | 364/365 selected tokens reused warm                                                                       |
| rotating-cache control    | controlled closed    | five seeds, four capacities                                                                               |
| native semantic parity    | controlled closed    | five seeds, zero logit error, 5/5 recovery                                                                |
| rotating plus native      | controlled closed    | 5/5 with 64-token local cache                                                                             |
| lazy native region        | prototype closed     | selected resources encode on first selection                                                              |
| consumer profile          | open                 | last 14/28 retains 5/5 on the controlled task; no reduced 4B–32B natural-QA profile passes the smoke gate |
| persistence               | controlled closed    | strict fingerprint; zero-error reload in 7.2 ms mean                                                      |
| shared storage lifecycle  | controlled closed    | segmented WARM 90/90 exact across Qwen/Llama/Gemma; restart recovery; int8 remains gated                  |
| concurrency/sharing       | controlled closed    | five seeds at 1–8 requests; 108 MiB avoided at eight                                                      |
| natural transport         | controlled closed    | 80/80 native; disabled/shuffled near chance                                                               |
| natural QA transport      | controlled closed    | original QASPER/Hotpot/2Wiki answers with causal controls                                                 |
| routed natural QA         | controlled partial   | 60/60 consumption parity; recall@4 0.348–0.775; oracle gap open                                           |
| matched E0/E2 economics   | controlled closed    | 2,086/2,086 expanded outputs; all four E2/E0 cost ratios exceed one                                       |
| quantized PRA detail      | controlled candidate | per-head int8 halves residency; native dequantization before attention                                    |
| memory pressure           | controlled closed    | 3,145 requests across compact, long-session, and crossed tier/window pressure                             |
| SOURCE/WARM admission     | controlled closed    | WARM is slower than SOURCE on the measured small-record cohort; no positive break-even reuse              |
| large local-window Pareto | open                 | 2K/8K/32K/full require occupied long histories and are not inferred from cache limits                     |

### 14.1 Observability contract

MLX does not require a container-native telemetry stack. PRA wraps Metal/MLX timing and residency observations in standard spans and normalized Prometheus metrics while the engine remains on the macOS host; Collector, Prometheus, and Grafana may run in containers. The dashboard distinguishes selected tokens, native K/V bytes, HOT/WARM promotion, rotating-cache behavior, TTFT, and ITL. All reported performance qualification keeps detailed tracing disabled.

## 15 Falsification and Limitations

Native PRA-MLX is unnecessary if selected visible text plus ordinary MLX caching matches its quality, memory, and latency on realistic long-memory workloads. The server smoke makes that baseline competitive. Conversely, the rotating control shows that bounded sequential KV is not automatically a PRA archive.

The original profile and natural controls center on one small 4-bit Qwen model; Llama/Gemma lifecycle rows remain smaller mechanism replications. The initial server experiment has only five repeats per condition, so its p95/p99 are not reported; the later gateway load curve reports finite-wave tails but serializes one model runner. The natural controls contain real dataset text but an inserted fixed answer-code mechanism. The first end-task study uses original answers but oracle-scoped evidence. The M4 scaling calibrations reach dense 32B and a 30B-A3B MoE checkpoint, but use only 30 examples per Qwen model. Llama 3.1 8B adds a 60-example cross-family concat-transport replication, but not a cross-family segmented-layer curve. The corrected cost ladder has five seeds but two examples per dataset and seed; its bootstrap intervals are smoke uncertainty, not a production noninferiority analysis. It establishes cost convergence toward parity and rejects the tested reduced-layer profiles,

not broad family generalization. The long-context extension likewise has 15 examples per 8K model point and only three at 32K; 64K exceeds the pinned model limit. It measures the MLX allocator rather than whole-system wired memory and does not exercise swap as a benchmark condition. The routed extension now has 50 sampled examples per condition and dataset (49 unique on QASPER), a parameter-free lexical/hash index, and four selected candidates. Strict EM is confounded by verbose continuation. Int8 detail is restored before attention, so it saves retained bytes rather than active attention bytes. The learned-router cohort has only 16 held-out examples per dataset/model and does not include 2Wiki; no learned native Q/K index, continuous batching, or OOM frontier is measured. The residency sweeps use deterministic cyclic schedules rather than a measured production session distribution. The original matched benchmark reuses only 20 examples per dataset; the expanded selector-frozen confirmation increases that cohort but does not establish end-task quality beyond proving that E2 preserves its E0 input. The rotating intervention also changes effective local context and can perturb chat-template semantics.

## 16 Related Work

MLX targets efficient and flexible machine learning on Apple Silicon [1]. Prompt caching and rotating K/V optimize sequential state; cache quantization reduces its physical size. Retrieval and context compression reduce visible input. PRA instead assigns stable identity to retained non-prefix memory and separates cheap addressing from active native detail. The mechanisms can be combined, but none is a semantic substitute for the others.

## 17 Conclusion

MLX prompt caching and selected context compose cleanly: the fixed answer is preserved with 76.9% fewer visible tokens, and repeated selected prompts reuse 364 of 365 tokens. Rotating K/V provides large layer-cache savings but fails as an archive in the five-seed control; late selected text does not reliably repair it. The in-process native executor passes the missing mechanism gate: selected non-prefix K/V is bit-exact to ordinary split-cache execution and preserves 5/5 recovery with a rotating local cache. PRA-MLX is therefore a working native prototype. Full-precision native transport also matches ordinary split-cache quality on original-answer QA, while wrong and absent memory fail causally. Per-head int8 halves retained detail without measured degradation in the small cohort. The M4 Pro replication extends this result to Qwen3-4B: ordinary and native aggregate F1 remain equal in all six cohorts and shuffled memory degrades sharply. The earlier 44.5–54.5% completion ratio is retired because it mixed lazy cold E0 with synchronized warm E2. The corrected 4B–32B ladder finds exact concatenated E2 quality, warm cost converging from 1.054 to 1.011 times E0, and a cold advantage contracting from 0.932 to 0.976. Full segmented overhead falls from 1.165 to 1.047, but no reduced consumer suffix passes the natural-QA quality gate. Model scale alone therefore supports cost convergence, not a native speed advantage or a smaller BALANCED consumer fraction. A separate 30-example-per-model M5 replication reaches a 0.980 concat point estimate and 1.031 segmented ratio at 14B, but the concat interval includes parity and 32B exceeds the 16 GiB host’s capacity gate. Its 1,500-request pressure extension reproduces the eight-resource residency threshold and separates repeat-load savings from cold p95. Across a 1,020-request hard-budget sweep, sub-working-set budgets cap compact residency but reload every repeated access; fitting all eight resources removes reloads and halves resolution time without changing F1. With SDK routing, native and ordinary paths remain identical on 60/60 paired generations, but recall@4 ranges from 0.348 to 0.775 and answer likelihood remains below oracle evidence. A five-seed router qualification improves QASPER recall across four MLX identities, but its HotpotQA transfer is negative, so generic routing remains the product default. The next quality improvement still belongs in discovery, not the MLX cache transport. Queued server pressure, an OOM frontier, broader learned-routing objectives, and larger quality cohorts remain necessary before production claims. Matched E2 execution removes roughly 91–93% of visible tokens and preserves every output, but its unfused decode is slower. Last-half consumption remains a narrow controlled result and int8 residency remains a candidate; neither is promoted by the natural scaling ladder. A segmented one-softmax primitive now replaces the live Qwen attention path and removes K/V concatenation. It retains a 5–17% completion-time overhead and small numerical drift because it is composed from unfused MLX operations; it is not a fused-kernel speed claim.

## References

- [1] A. Hannun et al. MLX: Efficient and Flexible Machine Learning on Apple Silicon. 2023.